



Quantum Cellular Automata and Categorical Dualities of Spin Chains

Corey Jones¹, Kylan Schatz¹, Dominic J. Williamson²

¹ Department of Mathematics, North Carolina State University, Raleigh, NC 27695, USA.

² School of Physics, University of Sydney, Sydney, NSW 2006, Australia.

E-mail: dominic.williamson@sydney.edu.au

Received: 5 May 2025 / Accepted: 22 January 2026

© The Author(s) 2026

Abstract: Dualities play a central role in the study of quantum spin chains, providing insight into the structure of quantum phase diagrams and phase transitions. In this work, we study categorical dualities, which are defined as bounded-spread isomorphisms between algebras of symmetry-respecting local operators on a spin chain. We consider generalized global symmetries that correspond to unitary fusion categories, which are represented by matrix-product operator algebras. A fundamental question about dualities is whether they can be extended to quantum cellular automata on the larger algebra generated by all local operators in the unit matrix-product operator sector. For on-site representations of Hopf algebra symmetries, this larger algebra is the usual tensor product quasi-local algebra. We present a solution to the extension problem using the machinery of Doplicher–Haag–Roberts bimodules. Our solution provides a crisp categorical criterion for when an extension of a duality exists. We show that the set of possible extensions form a torsor over the invertible objects in the relevant symmetry category. As a corollary, we obtain a classification result concerning dualities in the group case.

Contents

1.	Introduction	
2.	Dualities for Spin Chains with Categorical Symmetry	
3.	Abstract Fusion Spin Chains	
4.	The Algebra Model for Categorical Inclusions	
5.	Solving the Extension Problem with DHR Bimodules	
6.	Examples and Applications	

1. Introduction

Quantum cellular automata (QCA) are a class of quantum operations that capture the fundamental properties of unitarity and locality [Wat95,Far20,Arr19]. The problem of

classifying QCA up to finite depth unitary circuits has been the focus of much attention [GNVW09, SW04, FHH19, FH20, Haa21, Haa22], due to both the fundamental nature of QCA and their wide ranging applications. These include invertible quantum phases of matter [FHH20, JM21], floquet quantum dynamics [GNP21, PFM+16, PFVP17, PVF18, PM17, ZL21, AHLM23], many-body localization [LE24], unitary tensor network operators [CPGSV17, SSBC18, PC20, PTSC21, RWW20], and simulations of quantum field theories [ANW11, FS20, BM20].

Thus far, most work has focused on QCAs embodied by bounded-spread automorphisms on the standard quasi-local operator algebra of a spin system, $A = \otimes_{\mathbb{Z}^n} M_d(\mathbb{C})$, formed by taking the tensor product of finite qudit degrees of freedom on the sites of a lattice. This has led to a full classification of QCAs in one spatial dimension up to finite-depth circuits [GNVW09] and partial classifications in higher spatial dimensions [FH20, Haa21, Haa22]. There is a closely related problem of understanding *dualities*, which in this context are mathematically described by bounded-spread isomorphisms between *subalgebras* of the usual quasi-local algebra that interchange quantum phases of matter. Questions in this direction have attracted much recent interest [AMF16, TW19, CCH+21, KOZ22, LDOV23, Jon24, JL24, BDSNY24, MLC24].

Dualities can arise naturally in the presence of a global symmetry $\mathcal{C} \curvearrowright A$, which may correspond to an action of a group, a (weak) Hopf algebra, or a fusion category. If a pair of Hamiltonians has symmetric local terms, then these lie in the algebra $A^{\mathcal{C}} \subseteq A$ of symmetric operators. In this work, we use the term *duality* to formally denote a bounded-spread isomorphism between subalgebras of symmetric operators $A^{\mathcal{C}}$ under some global symmetry which exchange Hamiltonians *but may not extend* to bounded-spread automorphisms of the whole quasi-local algebra A (QCA). The motivating example for this class of dualities is due to Kramers and Wannier, originally introduced in the context of classical statistical mechanics [KW41]. The bounded-spread automorphism implementing KW duality acts on the symmetric subalgebra of operators on a chain of qubits that commute with the global $\mathbb{Z}_2$ symmetry represented by $\prod_i \sigma_i^x$. The action of the Kramers-Wannier (KW) duality can be specified on a set of generators of this $\mathbb{Z}_2$ -symmetric operator algebra via the map $\sigma_i^x \mapsto \sigma_{i-1}^z \sigma_i^z$ and $\sigma_i^z \sigma_{i+1}^z \mapsto \sigma_i^x$.

In recent years, there has been significant interest in extending results from the setting of conventional global symmetries, which are represented by tensor products of single-site unitary operators, to categorical global symmetries, which are represented by matrix product operators (MPOs) [FFRS04, BMW+17, VBV+18, AMF16, TW19]. In this work, we study this flavor of duality in one spatial dimension. The resulting algebra generated by symmetric local operators (preserving the unit MPO sector) is called a *fusion spin chain*. Recently, the idea of KW duality has been generalized in this direction to categorical symmetries, where matrix-product operators have been found to implement dualities based on any invertible bimodule categories between fusion categories [AMF16, WBV17, AFM20, LFH+20, LDOV23]. This suggests a deep connection between categorical dualities and the algebraic structure of fusion categories.

A key feature of non-trivial dualities following the KW paradigm is that bounded-spread isomorphisms of the symmetric operators *should not be* locally extendable to the whole spin chain (or in the case of non-unital MPOs, to the local edge-restricted sector, which is our term for the Hilbert space that the MPO corresponding to the monoidal unit in the fusion category projects onto). We call such an extension a *spatial implementation*. Spatial implementations are themselves ordinary 1+1D QCAs in the case of group or Hopf algebra symmetries. More generally, spatial implementations are *edge-restricted*

QCA. These extensions lead naturally to the following question, whose answer sheds light on the relationship between QCA and categorical dualities:

Question 1.1. *When do bounded-spread isomorphisms between symmetric local operator algebras (i.e. categorical dualities) extend to QCA defined on the full (or edge-restricted) local operator algebras, and how are the possible extensions characterized?*

To answer this question, we first establish a precise reformulation in terms of *abstract fusion spin chains* $A(\mathcal{E}, X)$, built from a fusion category $\mathcal{E}$ and an object $X \in \mathcal{E}$, by defining local algebras as endomorphism algebras of tensor powers of X (see Sect. 3). A fusion category $\mathcal{C}$ acting on a spin chain via MPOs is specified by the data of an indecomposable right $\mathcal{C}$ -module category $\mathcal{M}$, and an object $X \in \mathcal{C}_{\mathcal{M}}^*$, where $\mathcal{C}_{\mathcal{M}}^*$ denotes the unitary fusion category of $\mathcal{C}$ -module endofunctors (see e.g. [EGNO15]). Then, the symmetric operator algebra is isomorphic to the fusion spin chain $A(\mathcal{C}_{\mathcal{M}}^*, X)$. If we view $\mathcal{M}$ as a left $\mathcal{C}_{\mathcal{M}}^*$ -module category, then rewriting $\mathcal{E} := \mathcal{C}_{\mathcal{M}}^*$ we can reformulate the data of a categorical symmetry as an abstract fusion spin chain $A(\mathcal{E}, X)$ together with an indecomposable left $\mathcal{E}$ -module category $\mathcal{M}$. The inclusion $A(\mathcal{E}, X) \hookrightarrow A(\mathcal{E}, X)_{\mathcal{M}}$ (see Definition 3.5) corresponds to the inclusion of symmetric operators into all (edge-restricted) local operators, which we call a *spatial realization* of the chain $A(\mathcal{E}, X)$. Question 1.1 can be given a precise mathematical formulation as follows:

Question 1.2. *Given a bounded-spread isomorphism $\alpha : A(\mathcal{C}, X) \rightarrow A(\mathcal{D}, Y)$ between abstract fusion spin chains, when does it extend to a QCA between spatial realizations $A(\mathcal{C}, X)_{\mathcal{M}} \rightarrow A(\mathcal{D}, Y)_{\mathcal{N}}$, and how are these extensions characterized?*

In this form, we can address the problem using the machinery of Doplicher–Haag–Roberts (DHR) bimodules, introduced in [Jon24] and generalizing the earlier work of [NS95]. Associated to an abstract spin system A is a braided $\mathcal{C}^*$ -tensor category $\text{DHR}(A)$, and associated to any bounded-spread isomorphism $\alpha : A \rightarrow B$ is a braided equivalence $\text{DHR}(\alpha) : \text{DHR}(A) \cong \text{DHR}(B)$. Intuitively, if we view an abstract spin chain as operators living at a cut boundary of a 2+1D topologically ordered spin system (as in [JNPW23]) then DHR bimodules capture the structure of the bulk topological order (see also [CW23, IW23]). For a fusion spin chain $A(\mathcal{C}, X)$, there is a braided equivalence $\text{DHR}(A(\mathcal{C}, X)) \cong \mathcal{Z}(\mathcal{C})$, where the latter denotes the Drinfeld center of $\mathcal{C}$. We now set up the statement of our main theorem. For any indecomposable $\mathcal{C}$ -module category $\mathcal{M}$, let $Z(\mathcal{M}) \in \mathcal{Z}(\mathcal{C})$ denote the canonically associated Lagrangian algebra [Dav10]. Then we have the following (which is a special case of Theorem 5.6):

Theorem 1.3. *Let $\alpha : A(\mathcal{C}, X) \rightarrow A(\mathcal{D}, Y)$ be a bounded-spread isomorphism between abstract fusion spin chains. Then for any indecomposable module categories $\mathcal{M}$ and $\mathcal{N}$ of $\mathcal{C}$ and $\mathcal{D}$ respectively, spatial implementations $A(\mathcal{C}, X)_{\mathcal{M}} \rightarrow A(\mathcal{D}, Y)_{\mathcal{N}}$ of α are in bijective correspondence with algebra isomorphisms $\text{DHR}(\alpha)(Z(\mathcal{M})) \cong Z(\mathcal{N})$ in $\mathcal{Z}(\mathcal{D})$. As a consequence:*

- (1) *If $\text{DHR}(\alpha)(Z(\mathcal{M}))$ is not isomorphic to $Z(\mathcal{N})$, then α has no spatial implementation.*
- (2) *If $\text{DHR}(\alpha)(Z(\mathcal{M})) \cong Z(\mathcal{N})$, then the spatial implementations of α form a torsor over $\text{Aut}(Z(\mathcal{M})) \cong \text{Inv}(\mathcal{C}_{\mathcal{M}}^*)$*

Above, $\text{Inv}(\mathcal{C}_{\mathcal{M}}^*)$ denotes the group of isomorphism classes of invertible objects in $\mathcal{E}$. For the Kramers–Wannier example described above, the fusion category is $\mathcal{C} = \mathcal{D} = \text{Rep}(\mathbb{Z}_2)$, X is the left regular representation of $\mathbb{Z}_2$, and the DHR category of the symmetric spin chain is the toric code modular tensor category $\mathcal{Z}(\text{Hilb}_{f.d.}(\mathbb{Z}_2))$. The module

category in question is the standard fiber functor, which corresponds to the electric Lagrangian $L = 1 \oplus e$. The induced action of the Kramer-Wannier automorphism α is the $e \leftrightarrow m$ swap symmetry, which does not preserve L . By the above theorem, α admits *no* QCA extension to the whole quasi-local algebra $A = \otimes_{\mathbb{Z}} M_2(\mathbb{C})$.

We also use our formalism to make contact with the literature concerning symmetric QCA (sQCA). In this setting, we consider a finite global symmetry group G acting faithfully by on-site unitaries in a self-dual representation. Then the symmetric operators A^G have DHR category equivalent to the Drinfeld center $\mathcal{Z}(\text{Rep}(G))$. We immediately recover an $H^2(G, U(1))$ index (c.f. [GSSC20]) as a consequence of our framework (see Theorem 6.3).

In [JL24], a numerical index for bounded spread automorphisms on fusion spin chains was introduced which generalizes the GNVW index [GNVW09]. Combining this index with the DHR action, we obtain a homomorphism $\text{Ind} \times \text{DHR}$ from the group of dualities on an abstract fusion spin chain over a fusion category $\mathcal{C}$ to $\mathbb{R}^\times \times \text{Aut}_{br}(\mathcal{Z}(\mathcal{C}))$.

For an on-site action of a finite group G , letting A^G denote the quasi-local algebra of symmetric operators, we have a braided equivalence $\text{DHR}(A^G) \cong \mathcal{Z}(\text{Rep}(G))$. Thus the image of $\text{Ind} \times \text{DHR}$ is in $\mathbb{R}^\times \times \text{Aut}_{br}(\mathcal{Z}(\text{Rep}(G)))$. The group of braided autoequivalences of the center of a group-like fusion category is well-studied for many classes of G , see for example [ENO10, MN18]. Recently, a complete classification of abelian G -dualities for on-site regular actions was obtained [MLC24]. Using our results above, we obtain a similar result which also applies to non-abelian groups.

Corollary 1.4. *For any finite group G , a pair of G -dualities in the regular representation differ by a locally symmetric finite depth circuit if and only if they have the same numerical index and induce the same braided monoidal natural isomorphism on $\mathcal{Z}(\text{Rep}(G))$.*

The above corollary completely classifies the equivalence of G -dualities, including which dualities are trivial. However, it falls short of providing a closed form description of the image of $\text{Ind} \times \text{DHR}$ in $\mathbb{R}^\times \times \text{Aut}_{br}(\mathcal{Z}(\text{Rep}(G)))$. We leave this question to future work.

Our results point to a number of directions for future work. First, what is the full classification of bounded-spread isomorphisms on local algebras for fusion spin chains? Using our results, we expect to be able to reduce this problem to the classification of edge-restricted QCAs in one spatial dimension, which is an open problem. Second, can our results be applied to characterize symmetries of topological orders in two dimensions via their action by dualities on the holographic boundary (for example, in the sense of [JNPW23])? Third, can techniques from the theory of operator algebras, subfactors, and fusion categories be applied to classify dualities and QCAs in higher spatial dimensions?

The remainder of this work is presented as follows. In Sect. 2 we introduce background material on categorical dualities of quantum spin chains. In Sect. 3 we describe dualities in the context of fusion spin chains. In Sect. 4 we describe categorical inclusions defined by module categories over algebra objects. In Sect. 5 we describe how DHR bimodules can be applied to solve the extension problem for dualities of quantum spin chains. In Sect. 6 we present examples that generalize the KW duality, and give our applications.

2. Dualities for Spin Chains with Categorical Symmetry

In this section we introduce relevant background material. First, we briefly recall the algebraic approach to spin chains [Naa17, BR97]. Consider $\mathbb{Z} \subseteq \mathbb{R}$ as a metric space. A spin chain is specified by choosing an on-site Hilbert space $\mathbb{C}^d$; for every interval

$I \subseteq \mathbb{Z}$, we have the local Hilbert space $\otimes_{x \in I} \mathbb{C}^d$ and the algebra of operators localized in I :

$$A_I := \otimes_{x \in I} M_d(\mathbb{C}).$$

For $I \subseteq J$, we have natural inclusions $A_I \hookrightarrow A_J$ defined by $a \mapsto 1^{J < I} \otimes a \otimes 1^{J > I}$. The C^* -completion of the union over all intervals (or more formally, the colimit in the category of $*$ -algebras) is called the *quasi-local algebra*. We call the *algebraic union* the local algebra. States of the system in the thermodynamic limit are normalized, positive linear functionals on A , while Hamiltonians can be expressed as (unbounded) derivations on A (see [BR87, BR97]).

A *local Hamiltonian* is an assignment of a self-adjoint element $H_I \in A_I$ to every interval $I \subseteq \mathbb{Z}$ for which there exists an $R \geq 0$ such that if I has length greater than R , then $H_I = 0$. This defines an (unbounded) derivation on the local algebra:

$$\delta_H(a) := \sum_I [H_I, a],$$

which is well-defined since the above summation contains only finitely many non-zero terms. We point out that what we call local Hamiltonians are often called *strictly local* in the literature, since we assume our interaction terms have finite range. Nevertheless, we will use the term local following [ZCZW19].

The primary motivation for this paper is to understand a class of *categorical dualities* between spin systems. Broadly speaking, a duality between two physical theories is a mapping between the observables of the theories that intertwines the relevant physics (state, dynamics, etc.). However, interesting dualities should be equivalences that are in some salient sense “not obvious”. In the case of spin systems dualities are generally considered in the context of symmetries, whereby we have a locality-preserving equivalence on the symmetric subsystems that does not extend to an equivalence of the whole theory (for example, see [ECN11] or for the categorical context [LDOV23, AMF16]). We give a mathematical formulation of this in 1D in terms of fusion spin chains.

First we set up some basic terminology. We need a natural notion of a locality-preserving equivalence of spin systems, which is captured by the idea of bounded-spread isomorphisms (or quantum cellular automata).

Definition 2.1. If A, B are the local algebras of spin chains on $\mathbb{Z}$, a *bounded-spread isomorphism* is a $*$ -isomorphism $\alpha : A \rightarrow B$ such that there exists an $R > 0$ so that $\alpha(A_I) \subseteq B_{I+R}$ for every interval I . A bounded-spread isomorphism $\alpha : A \rightarrow A$ is called a *quantum cellular automaton* (QCA).

In the above definition $I+R$ denotes the R -neighborhood of the interval I in $\mathbb{Z}$. A bounded-spread isomorphism is a natural mapping between the observables of the two spin chains that *uniformly* preserves locality.

Definition 2.2. If H, K are local Hamiltonians on spin chains with local algebras A and B respectively, an equivalence between them is a bounded-spread isomorphism $\alpha : A \rightarrow B$ such that $\alpha(\delta_H(\alpha^{-1}(b))) = \delta_K(b)$ for all local operators $b \in B$. We denote this by $\alpha(H) = K$.

By the definition of bounded-spread isomorphism, for any local Hamiltonian $H = \{H_I\}$ on A , $\alpha(H) := \{\alpha(H)_I := \alpha(H_{I-R})\}$ is a local Hamiltonian on B and α implements an equivalence between these. Thus given any bounded-spread isomorphism $\alpha : A \rightarrow B$ and any local Hamiltonian H on A , α always implements an equivalence between H and *some* local Hamiltonian on B . This suggests that it could be useful to shift our focus to the groupoid of local algebras of spin chains and bounded-spread isomorphisms between them.

Example: Finite depth quantum circuits. An obvious class of QCA are the *finite-depth quantum circuits*, (abbreviated FDQC). Recall a depth-one circuit is given by specifying a uniformly bounded partition $\mathcal{P}$ of $\mathbb{Z}$ into intervals, and for $I \in \mathcal{P}$, choosing a unitary $u_I \in A_I$. Then define a QCA acting on the local operator x by

$$\alpha(x) := \left(\prod_{I \in \mathcal{P}} u_I \right) x \left(\prod_{I \in \mathcal{P}} u_I^* \right).$$

Note that by locality, this infinite product is in fact finite since all but finite many of the u_I commute with x .

A *finite-depth quantum circuit* (FDQC) is a finite composition of depth one circuits. FDQC are QCA that can be locally implemented. It is easy to show that FDQC is a normal subgroup of QCA. FDQC equivalence for states is used to characterize long-range entanglement classes [CGW10], and thus we can view QCA/FDQC as long-range entanglement classes of uniformly locality-preserving equivalences [Haa22]. Since FDQC are locally implemented, they are often considered “trivial” as equivalences of Hamiltonians, which naturally leads us to consider the group $QCA/FDQC$, which has been frequently studied in the literature.

Before moving on to giving a mathematical definition for our Kramers-Wannier type dualities, it is useful to have in hand the notion of an *abstract spin chain*. This is strongly motivated by the Haag-Kastler axioms for an algebraic quantum field theory [HK64], and is a straightforward generalization of the properties satisfied by the quasi-local algebra of a spin system [BR97].

Definition 2.3. An abstract spin chain is a unital C^* -algebra A and for every interval I , a unital subalgebra $A_I \subseteq A$ such that

- (1) (Isotony) If $I \subseteq J$, $A_I \subseteq A_J$
- (2) (Locality) If $I \cap J = \emptyset$, then $[A_I, A_J] = 0$
- (3) (Density) $(\cup A_I)^{\|\cdot\|} = A$
- (4) (Weak algebraic Haag duality) There is some $R \geq 0$ such that for every interval I , $\{x \in A \mid [x, y] = 0 \text{ for all } y \in A_J, J \subseteq I^c\} \subseteq A_{I+R}$.

If the last condition is satisfied with $R = 0$, it is called *algebraic Haag duality*. The abstract spin chains of interest in this paper always satisfy this stronger version of the last axiom. For two abstract spin chains, the definition of *bounded-spread isomorphism* used above makes perfect sense unchanged. We note that if α is bounded spread, then it follows from weak algebraic Haag duality that α^{-1} is as well.

Actions of fusion categories on spin chains. We are now ready to describe dualities, in the same sense as [LDOV23]. Suppose we have two spin systems with local algebras A and B , both with a global symmetry $\mathcal{C} \hookrightarrow A$ and $\mathcal{D} \hookrightarrow B$ (think of an on-site group symmetry, or more generally, a categorical symmetry determined by an MPO, see below). Suppose further that the local terms of the Hamiltonians H and K are both symmetric. Then we can restrict our attention to the symmetric operators, which consist of subalgebras $A^{\mathcal{C}} \subseteq A$ and $B^{\mathcal{D}} \subseteq B$ containing the terms of H and K respectively. We can then ask for bounded-spread isomorphisms *between the subalgebras* $A^{\mathcal{C}}$ and $B^{\mathcal{D}}$ which intertwine H and K . This is what we call a duality.

First, we formalize what we mean by a fusion categorical symmetry and the algebras of symmetric operators, resulting in the definition of fusion spin chains and duality. Recall a unitary fusion category is a finitely semi-simple rigid C^* -tensor category with simple unit object (see e.g. [NT13, LR97]). Recall that if $\mathcal{C}$ is a fusion category, an object $X \in \mathcal{C}$ is *strongly tensor generating* if there exists some $n \geq 0$ such that every simple object in $\mathcal{C}$ is isomorphic to a summand of $X^{\otimes n}$. We emphasize that this n needs to be independent of the simple object in question, which distinguishes this notion from the usual definition of tensor generating. For further definitions concerning fusion categories, module categories, and related concepts, see [EGNO15].

Definition 2.4. Let A be a spin system with local Hilbert space $\mathbb{C}^d$ and let $\mathcal{C}$ be a unitary fusion category. An action of $\mathcal{C}$ on A consists of

- (1) An indecomposable, semi-simple right module category $\mathcal{M}$ of $\mathcal{C}$.
- (2) An object $X \in \mathcal{C}_{\mathcal{M}}^*$ which is *strongly tensor generating* for $\mathcal{C}_{\mathcal{M}}^*$
- (3) A unitary isomorphism $\bigoplus_{i,j \in \text{Irr}(\mathcal{M})} \mathcal{M}(i, X \triangleright j) \cong \mathbb{C}^d$.

The action is called *unital* if $\mathcal{M}$ is equivalent to $\text{Hilb}_{f,d}$ as C^* -categories.

Remark 2.5. In the above definition, we need to equip the morphism spaces $\mathcal{M}(i, X \triangleright j)$ with the structure of a Hilbert space. The most general way to do this is pick a *module trace* (see [GMP+23]), but for our purposes the choice of module trace is irrelevant since any distinct choices give rise to unitarily equivalent Hilbert spaces. We will use here (and in the rest of the paper) the *composition inner product*: given $f, g \in \mathcal{M}(i, X \triangleright j)$, define the composition inner product by $\langle f | g \rangle := f^* \circ g$.

From the data of an action, we can construct *matrix product operators* which act by quantum channels on (a sector of) the spin system, implementing the fusion rules of $\mathcal{C}$.

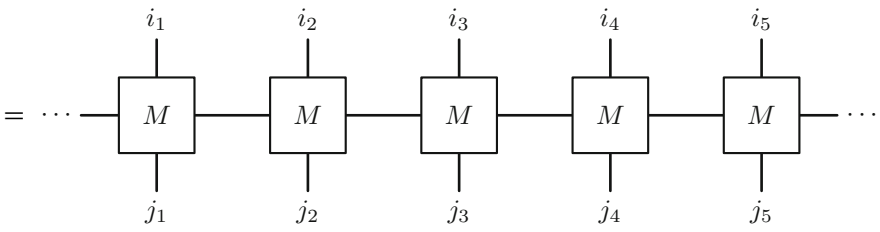
In general, a matrix product operator is defined from a 4-valent tensor

$$M_{\alpha\beta}^{ij} := \alpha \text{ --- } \boxed{M} \text{ --- } \beta \text{ ,} \quad \begin{array}{c} i \\ \updownarrow \\ j \end{array} \quad (2.6)$$

in the above equation the diagram is internal to $\mathbf{Vec}$ and the external lines carry finite dimensional vector spaces $\mathbb{C}^d$. The full matrix product operator is

$$\sum_{\{i\}\{j\}} \text{tr}(M^{i_1 j_1} M^{i_2 j_2} \dots M^{i_n j_n}) |\{i\}\rangle \langle \{j\}| \quad (2.7)$$

where $\{i\} = \{i_1, i_2, \dots, i_n\}$ and similarly for $\{j\}$, and

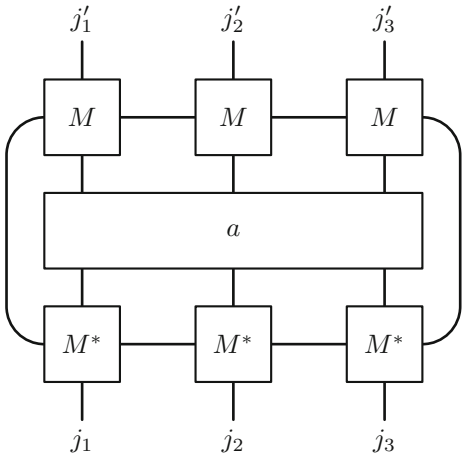
$$\text{tr}(M^{i_1 j_1} M^{i_2 j_2} \dots M^{i_n j_n})$$

(2.8)

where we treat $M_{\alpha\beta}^{ij}$ as a matrix element of M^{ij} and take a matrix product over the α, β indices of the tensors $M_{\alpha\beta}^{ij}$. From the data in Definition 2.4, one can construct for each (simple) object in $a \in \mathcal{C}$ a matrix product operator, MPO_a . These matrix product operators form a representation of the fusion ring of $\mathcal{C}$,

$$\text{MPO}_a \text{MPO}_b = N_{a,b}^c \text{MPO}_c \quad (2.9)$$

and furthermore, the local tensor satisfy a family of categorical relations described in detail in Refs. [SWB+14, BMW+17]. It is in this sense that a fusion category is actually acting on the spin chain. Since MPOs are not strictly necessary to establish our results, we refer to references [LDOV23, Kaw20] for the details concerning the formal definition of MPO tensors from our initial data

Alternatively, one can build from an MPO a locality preserving quantum channel, which is defined on local operators via a suitable normalization of the following picture


(2.10)

where M^* denotes the complex conjugate of M with the i and j indices swapped. The categorical relations imply that these quantum channels implement a normalized version of the fusion rules of the category.

Edge-restricted algebra. For non-unital fusion categorical symmetries, it is natural to consider a non-unital “subalgebra” of A , called the *edge-restricted algebra*. If we pick an

orthonormal basis $E_{i,j}$ for the space $\mathcal{M}(i, X \triangleright j)$, we can define a finite graph $\mathcal{G}$ whose vertices are $\text{Irr}(\mathcal{M})$, and the edges from $a \rightarrow b$ are defined to be $E_{a,b}$. Then using the isomorphism between the onsite Hilbert space $\mathcal{H}_i$ and $\bigoplus_{i,j \in \text{Irr}(\mathcal{M})} \mathcal{M}(i, X \triangleright j) \cong \mathbb{C}^d$, we have a distinguished basis of $\mathcal{H}_m$ indexed by the edges of $\mathcal{G}$, for every $m \in \mathbb{Z}$.

Then for each pair of adjacent sites $(i, i+1)$, consider the local projection $P_i \in A_{[i,i+1]}$ defined by $P_i(e \otimes f) = \delta_{t(e)=s(f)} e \otimes f$, which checks the adjacency of the edges e and f in the graph $\mathcal{G}$. Note that $[P_i, P_j] = 0$ for all i and j , and thus we can (unambiguously) consider the projection

$$P := \prod_{i \in \mathbb{Z}} P_i.$$

We have two immediate observations:

- (1) This projection in general does not lie in the quasi-local C*-algebra A . However, if we pick a state ϕ on A such that $\phi(P_i \cdot P_i) = \phi$ for all i , then P does define an operator on the GNS Hilbert space $L^2(A, \phi)$ in the von Neumann completion of A . If we wish to avoid picking a state, we can view P as an element of the von Neumann algebra A^{**} , which is the von Neumann completion of A in the universal Hilbert space representation of A [Tak79].
- (2) If we consider a finite chain on an interval I , then the action of the restricted version of P , P_I on the Hilbert space associated to I , $\mathcal{H}_I$, agrees with the action of the MPO assigned to the unit object $\mathbb{1} \in \mathcal{C}$ away from the boundary. For this reason, it is reasonable to call the image of P in a Hilbert space representation the *unit MPO sector*. This is the sector on which the categorical MPO operators satisfy the required categorical relations [BMW+17].

We can consider the “corner subalgebra” of A , defined by $A^\circ := PAP$ (in A^{**}), with local subalgebras $A_I^\circ := PA_I P$. Technically speaking, the the elements A_I° are not properly elements of the algebra A or even its C*-completion, but rather they are elements of the von Neumann completion of A (either in the double dual von Neumann algebra A^{**} , or in any faithful Hilbert space representation of A where P does not act by 0).

We can easily identify A_I° as follows: If $I = [a, b]$ with $b - a + 1 = n$, let $L_I^{i,j}$ be the Hilbert space with orthonormal basis given by the paths of length n in $\mathcal{G}$ starting with element i and ending with element j . Let $d_{i,j}^n$ denote the number of length n paths from i to j so that $d_n^{i,j} = \dim(L_I^{i,j})$.

Then we have an isomorphism of Hilbert spaces to

$$\left(\prod_{a \leq i < b} P_i \right) (\mathcal{H}_I) \cong \bigoplus_{i,j \in \mathcal{G}} L_I^{i,j}.$$

It is easy to see from the definitions that

$$A_I^\circ \cong \bigoplus_{i,j \in \text{Irr}(\mathcal{M})} \text{Mat}(L_I^{i,j})$$

where the summand $\text{Mat}_{d_n^{i,j}}(\mathbb{C})$ is identified with all the linear operators on the Hilbert space $L_J^{i,j}$. In other words, A_I° is isomorphic to the subalgebra of operators on $\mathcal{H}_I$ that satisfy the property that the coefficient of the matrix unit $e_{p,q}$ is 0 if $s(p) \neq s(q)$ or $t(p) \neq t(q)$.

Note that from this description, A_I° is isomorphic to a (non-unital) subalgebra of A_I , but not coherently. Nevertheless, A° has the structure of an abstract spin chain. All of the axioms are obvious except for weak algebraic Haag duality, which in fact follows from Lemma 4.3 below.

Symmetric algebra. The operators of interest to us are the symmetric local operators. To define these, for each $Y \in \text{Irr}(\mathcal{C})$, and interval J , let M_J^Y denote the associated matrix product operator, acting on (a sufficiently large) edge-restricted Hilbert space with periodic boundary conditions

$$L_J^{per} := \bigoplus_{i \in \text{Irr}(\mathcal{M})} L_J^{i,i},$$

where the sum over i , i reflects the periodic boundary condition. Note that the matrix product operator maps the edge-restricted Hilbert space with periodic boundary conditions back to itself [BMW+17].

If I and J are intervals with $I \subseteq J$, then we say $I \ll J$ (read J is sufficiently larger than I) if the number of vertices in J to the left (and respectively right) of I , denoted n_L (and n_R , respectively) are large enough so that any two vertices in $\mathcal{G}$ are connected by a path of length n_L and n_R . This follows from the fact that $\mathcal{M}$ is indecomposable and X is strongly tensor generating for $X \in \mathcal{C}_{\mathcal{M}}^*$. Indeed, if n is such that all simple objects in $\mathcal{C}_{\mathcal{M}}^*$ are isomorphic to a summand of $X^{\otimes n}$, then it suffices to require $n_L, n_R \geq n$. Then for any interval $I \ll J$, we have a unital, faithful representation $A_I^\circ \subseteq \text{Mat}(L_J^{per})$. Then we define

$$A_I^{\mathcal{C}} := \{x \in A_I^\circ \mid [M_J^Y, x] = 0\}.$$

Notice that since the MPO are defined locally, this does not depend on the choice of ambient interval $I \ll J$.

For $I \subseteq J$, we have natural inclusions $A_I^{\mathcal{C}} \hookrightarrow A_J^{\mathcal{C}}$, and taking the colimit in the category of C^* -algebras, we obtain the algebra $A^{\mathcal{C}}$. Identifying $A_I^{\mathcal{C}}$ with its image in the colimit gives an abstract spin system. We call the algebra $A^{\mathcal{C}}$ the *symmetric algebra*.

We can now formally define a version of duality for Hamiltonians generalizing Kramers-Wannier duality to the categorical setting, following [LDOV23]. We can then state the problem that motivates this work.

Definition 2.11. Let $H = \{H_I\}$ be a local Hamiltonian on the spin system A , and let $\mathcal{C}$ be a fusion category acting on A . Then we say H is $\mathcal{C}$ -symmetric if for every interval I with $H_I \neq 0$, $H_I P = P H_I \neq 0$ and $H_I P \in A_I^{\mathcal{C}}$.

In this case the collection $H^\circ := \{H_I P\}$ actually defines a local Hamiltonian internally to the symmetric chain $A^{\mathcal{C}}$, which we denote $H^{\mathcal{C}} := \{H_I^{\mathcal{C}}\}$

Definition 2.12. Let A and B be spin chains with local Hamiltonians H and K respectively. A *categorical duality* between (A, K) and (B, K) consists of:

- (1) Categorical symmetries $\mathcal{C} \curvearrowright A$, $\mathcal{D} \curvearrowright B$ such that H is $\mathcal{C}$ -symmetric and K is $\mathcal{D}$ -symmetric

- (2) A bounded-spread isomorphism $\alpha : A^{\mathcal{C}} \rightarrow B^{\mathcal{D}}$ such that $\alpha(H^{\mathcal{C}}) = K^{\mathcal{D}}$ (in the sense of Definition 2.2, but restricted to $A^{\mathcal{C}}$).

Motivating Example: Kramers-Wannier. Here we spell out the motivating example of Kramers-Wannier duality [KW41]. In this case, the acting fusion category is $\mathcal{C} = \text{Hilb}_{f.d.}(\mathbb{Z}_2)$. The module category is $\mathcal{M} = \text{Hilb}_{f.d.}$ equipped with the trivial $\text{Hilb}_{f.d.}(\mathbb{Z}_2)$ action, and the dual object $X = \mathbb{C}[\mathbb{Z}_2] \in \text{Rep}(\mathbb{Z}_2)$ is the left regular representation.

This translates into the standard on-site spin flip action of $\mathbb{Z}_2$ on a chain of qubits $\otimes_{\mathbb{Z}} \mathbb{C}^2$, so that $A = \otimes_{\mathbb{Z}} M_2(\mathbb{C})$. The non-trivial symmetry operator (or MPO) can be represented by the product of on-site operators (i.e. with trivial valence bonds) as

$$U_g := \prod \sigma_i^x.$$

In this case, since $\mathcal{M}$ is rank one, the action is unital, so that the edge-restricted algebra and the quasi-local algebra of the entire spin system coincide, $A^\circ = A$.

The symmetric algebra is the C*-algebra generated by the operators σ_i^x and $\sigma_i^z \sigma_{i+1}^z$ as i ranges over $\mathbb{Z}$, or more formally

$$A^{\mathbb{Z}_2} = C^* - \text{alg}\{\sigma_i^x, \sigma_i^z \sigma_{i+1}^z \mid i \in \mathbb{Z}\}.$$

Then consider the Ising Hamiltonian

$$H_{J,h} = \sum_{i \in \mathbb{Z}} J \sigma_i^z \sigma_{i+1}^z + h \sigma_i^x,$$

which is $\mathbb{Z}_2$ -symmetric according to our definition, since all local terms lie in the subalgebra $A^{\mathbb{Z}_2}$.

Now, the Kramers-Wannier duality is defined by

$$\alpha(\sigma_i^x) = \sigma_{i-1}^z \sigma_i^z \text{ and } \alpha(\sigma_i^z \sigma_{i+1}^z) = \sigma_i^x,$$

which extends to an isomorphism $\alpha : A^{\mathbb{Z}_2} \rightarrow A^{\mathbb{Z}_2}$. We see immediately from the definition

$$\alpha(H_{J,h}) = H_{h,J}.$$

A key feature of KW-duality is that the equivalence between theories *can only be witnessed* at the level of symmetric operators, and does not extend to a (uniformly locality-preserving) equivalence of the whole spin system [ECN11]. Alternatively, some authors say that any unitary extension does not preserve locality, though this is not rigorously defined in the algebraic formalism for the thermodynamic limit we are using here.

This leads us to the following definition.

Definition 2.13. A bounded-spread isomorphism $\alpha : A^{\mathcal{C}} \rightarrow B^{\mathcal{D}}$ is called *spatially implemented* if it admits an extension to a bounded-spread isomorphism $\tilde{\alpha} : A^\circ \rightarrow B^\circ$.

The idea is that spatially implemented bounded-spread isomorphisms are not proper dualities in the sense of KW, and are actually more of a standard equivalence. This is not literally true on the nose in full generality, since the $\tilde{\alpha} : A^\circ \rightarrow B^\circ$ generally does not extend to $A \rightarrow B$, and thus more properly we would say that $\tilde{\alpha} : A^\circ \rightarrow B^\circ$ such

that $\tilde{\alpha}(H^{\mathcal{C}}) = K^{\mathcal{D}}$ is an *emergent equivalence* [ECN11]. Some may view an emergent equivalence as a kind of duality, but for us we regard it as closer to an equivalence than a KW-type duality. Indeed, if $A^{\circ} = A$ and $B^{\circ} = B$ (or in other words, the categorical action is unital, or equivalently, the module categories correspond to fiber functors), then the emergent equivalence reduces to an ordinary equivalence.

In any case, we can now give a precise mathematical statement of Question 1.1:

Reformulation of Question 1.1. Given a bounded-spread isomorphism $\alpha : A^{\mathcal{C}} \rightarrow B^{\mathcal{D}}$, can we characterize when it is spatially implemented?

In the sections that follow, we present a precise and abstract statement of this problem, and a solution using the DHR bimodule category of fusion spin chains.

3. Abstract Fusion Spin Chains

In the previous section, we set up the problem of studying bounded-spread isomorphisms between algebras of operators invariant under a categorical symmetry (i.e. $A^{\mathcal{C}}$), and asked when this can be extended to the edge-restricted algebra (i.e. A°). In this section, we present an abstract formulation of this problem in terms of fusion spin chains. We refer the reader to [EGNO15] for definitions of fusion categories and related objects. For convenience, we assume all our fusion categories are strict.

Fusion spin chains. Given an indecomposable multi-fusion category $\mathcal{E}$ and an object $X \in \mathcal{E}$ (not necessarily simple) we construct an abstract spin chain $A(\mathcal{E}, X)$ as follows:

- For each interval I , define $A(\mathcal{C}, X)_I := \mathcal{E}(X^{\otimes |I|}, X^{\otimes |I|})$
- If $I \subseteq J$, then we define the inclusion $A(\mathcal{E}, X)_I \hookrightarrow A(\mathcal{E}, X)_J$ by

$$f \mapsto (1_X)^{\otimes J < I} \otimes f \otimes (1_X)^{\otimes J > I}.$$

- We set $A(\mathcal{E}, X) = \operatorname{colim}_I A(\mathcal{E}, X)_I$, where the colimit is taken in the category of $*$ -algebras.

The commutativity property follows from functoriality of the monoidal product. Abstract spin chains constructed as above are called *fusion spin chains* [Jon24, JL24].

It turns out the three examples of abstract spin chains we saw in the previous section can be realized as fusion spin chains:

- (1) **Concrete spin systems.** Choose the fusion category $\mathcal{E} = \operatorname{Hilb}_{f.d.}$ of finite-dimensional Hilbert spaces, and pick the object $X := \mathbb{C}^d$. Then $A(\operatorname{Hilb}_{f.d.}, X)$ can be identified with the ordinary spin chain A constructed from $\mathbb{C}^d$.
- (2) **Edge-restricted algebras.** Consider a fusion category action on an ordinary spin system $\mathcal{C} \curvearrowright A$, and let $X \in \mathcal{C}_{\mathcal{M}}^*$ be the choice of strong tensor generator. This defines an object in the multi-fusion category $\tilde{X} \in \mathcal{E} = \operatorname{Mat}_m(\operatorname{Hilb}_{f.d.})$, where m is the rank of $\mathcal{M}$. This is the multi-fusion category consisting of $m \times m$ matrices of finite-dimensional Hilbert spaces, with $\otimes$ given by “matrix-multiplication” [FHP20]. The object $\tilde{X}$ is defined by

$$\tilde{X}_{i,j} := \mathcal{M}(i, X \triangleright j)$$

with Hilbert space structure on the components given by the composition inner product. It is easy to see that the edge-restricted abstract spin chain A° associated with our action $\mathcal{E} \curvearrowright A$ can be identified with the fusion spin chain $A(\operatorname{Mat}_m(\operatorname{Hilb}_{f.d.}), \tilde{X})$.

- (3) **Symmetric algebras** Again, suppose we have a fusion category $\mathcal{C} \curvearrowright A$. Then we can identify $A^{\mathcal{C}}$ with the fusion spin chain $A(\mathcal{C}_{\mathcal{M}}^*, X)$ [Kaw22, Kaw21].

This leads us to define the following groupoid:

Definition 3.1. Dua is the groupoid whose

- (1) Objects are fusion spin chains $A(\mathcal{E}, X)$.
- (2) Morphisms are bounded-spread isomorphisms with composition.

Our discussion above has motivated the definition of Dua, whose objects are *abstract* spin systems of symmetric operators. However, we are interested in studying spin systems *together with an embedding into an edge-restricted algebra* for some categorical symmetry. Here we give an abstract formulation. First recall that a functor $F : \mathcal{E} \rightarrow \mathcal{D}$ is *dominant* if every object in $\mathcal{D}$ is isomorphic to a sub-object of some $F(E)$, with $E \in \mathcal{E}$

Definition 3.2. Let $\mathcal{E}$ be an indecomposable multi-fusion category. A *quotient* of $\mathcal{E}$ is an indecomposable multifusion category $\mathcal{D}$ and a dominant tensor functor $F : \mathcal{E} \rightarrow \mathcal{D}$

We typically consider quotients up to (unitary) monoidal equivalence. Now, given any dominant tensor functor $F : \mathcal{E} \rightarrow \mathcal{D}$, we have a natural inclusion of nets,

$$A(\mathcal{E}, X) \hookrightarrow_{\iota_F} A(\mathcal{D}, F(X)),$$

where $f \in A(\mathcal{E}, X)_I = \mathcal{E}(X^{\otimes n}, X^{\otimes n})$ is sent to the composition

$$F(X)^{\otimes n} \xrightarrow{\text{can}} F(X^{\otimes n}) \xrightarrow{F(f)} F(X^{\otimes n}) \xrightarrow{\text{can}} F(X)^{\otimes n},$$

in $A(\mathcal{D}, F(X))_I$. Here *can* denotes the canonical unitary isomorphisms built from the tensorators on F . The map ι_F is a unital inclusion of the local C^* -algebras for each I , and is compatible with the connecting inclusions associated with $I \subseteq J$. Thus ι_F extends to a unital inclusion of quasi-local algebras. If we have a quotient $F : \mathcal{C} \rightarrow \mathcal{D}$ and a quotient $G : \mathcal{D} \rightarrow \mathcal{E}$, their composition $G \circ F : \mathcal{C} \rightarrow \mathcal{E}$ is a quotient. Furthermore, it is easy to see by construction that $\iota_{G \circ F} := \iota_G \circ \iota_F$. We call such an inclusion of nets a *categorical inclusion*.

Remark 3.3. We note that dominance is not required to get an inclusion of algebras ι_F as above. Indeed, any unitary tensor functor will do. However, we need dominance in the sequel to guarantee that if X is a strong tensor generator for $\mathcal{E}$, then $F(X)$ is a strong tensor generator for $\mathcal{D}$, which will be important from the perspective of categorical symmetries.

Given a finitely semi-simple category $\mathcal{M}$, its category of endofunctors $\text{End}(\mathcal{M})$ is an indecomposable multi-fusion category equivalent to $\text{Mat}_n(\text{Hilb}_{f.d.})$, where $n = \text{rank}(\mathcal{M})$. Recall that the data of a (left) $\mathcal{E}$ -module category structure on a finitely semi-simple category $\mathcal{M}$ can be expressed as a tensor functor $F : \mathcal{E} \rightarrow \text{End}(\mathcal{M})$, where $F(Y) := Y \triangleright \cdot$.

Lemma 3.4. *If $\mathcal{M}$ is an indecomposable left $\mathcal{E}$ -module category then the unitary tensor functor $F : \mathcal{E} \rightarrow \text{End}(\mathcal{M})$ is dominant.*

Proof. Suppose $\mathcal{M}$ is indecomposable and consider $W = \bigoplus_{Y \in \text{Irr}(\mathcal{E})} Y$, and consider $F(W) := W \triangleright \cdot \in \text{End}(\mathcal{M})$. Then we claim that for any simple objects $m, n \in \mathcal{M}$, $\mathcal{M}(m, W \triangleright n) \neq 0$. Indeed, suppose for contradiction that for some m, n , $\mathcal{M}(m, W \triangleright n) = 0$. Then for all $Y \in \text{Irr}(\mathcal{E})$, $\mathcal{M}(m, Y \triangleright n) = 0$, and thus

$$\mathcal{M}(m, \mathcal{E} \triangleright n) = 0.$$

Hence the idempotent completion of $\mathcal{E} \triangleright n$ is a full $\mathcal{E}$ -module subcategory of $\mathcal{M}$, hence a summand (see, for example, [EGNO15, Proposition 7.6.7]). But this is a proper module subcategory since m is disjoint from it, contradicting indecomposability. Now since $\mathcal{M}(m, W \triangleright n) \neq 0$, any object $G \in \text{End}(\mathcal{M})$ can be realized as a summand of $F(W^{\oplus k})$, where $k = \max_{m, n \in \text{Irr}(\mathcal{M})} \{\dim(\mathcal{M}(m, G(n)))\}$. Thus F is dominant. $\square$

Definition 3.5. A *spatial realization* of a fusion categorical net $A(\mathcal{E}, X)$ is a categorical inclusion $A(\mathcal{E}, X) \hookrightarrow A(\text{End}(\mathcal{M}), X \triangleright \cdot)$ associated to an indecomposable left $\mathcal{E}$ -module category $\mathcal{M}$. We denote $A(\text{End}(\mathcal{M}), X \triangleright \cdot)$ by $A(\mathcal{E}, X)_{\mathcal{M}}$.

We note that when $\text{rank}(\mathcal{M}) = 1$, $\text{End}(\mathcal{M}) \cong \text{Hilb}_{f.d.}$, and hence a spatial realization is an honest unital inclusion of a fusion spin chain into the local operators of a concrete spin chain.

The motivation for the above definition arises from our concept of fusion categorical symmetries. These symmetries are characterized by the following data: a fusion category $\mathcal{C}$, an indecomposable right module category $\mathcal{M}$, and a (strongly tensor generating) object $X \in \mathcal{C}_{\mathcal{M}}^*$. This data corresponds precisely to a spatial realization of the fusion categorical net $A(\mathcal{C}_{\mathcal{M}}^*, X)$ (up to a suitable notion of equivalence).

If we change our perspective and denote $\mathcal{E} := \mathcal{C}_{\mathcal{M}}^*$, then we can view $\mathcal{M}$ as an indecomposable left $\mathcal{E}$ -module category, and thus spatial realizations of the fusion spin chain $A(\mathcal{E}, X)$, parameterized by a left module category $\mathcal{M}$, correspond precisely to the algebra of symmetric operators inside A° , for an action of the dual category $\mathcal{E}_{\mathcal{M}}^* \curvearrowright A$. This leads us to reformulate Question 2 as follows:

Reformulation of Question 1.1. Let $A(\mathcal{E}, X)$ and $A(\mathcal{D}, Y)$ be fusion spin chains, with spatial realizations parameterized by module categories $\mathcal{M}$ and $\mathcal{N}$ respectively. If $\alpha : A(\mathcal{E}, X) \rightarrow A(\mathcal{D}, Y)$ is a bounded-spread isomorphism, when does α extend to a bounded-spread isomorphism $\tilde{\alpha} : A(\mathcal{E}, X)_{\mathcal{M}} \rightarrow A(\mathcal{D}, Y)_{\mathcal{N}}$?

4. The Algebra Model for Categorical Inclusions

One way to build a categorical inclusion is in terms of connected, commutative Q-systems. Q-systems are a particular axiomatization of the notion of C*-Frobenius algebra object internal to a unitary tensor category. In particular, if $\mathcal{A}$ is a unitary tensor category, a Q-system (following the definition conventions of [CHJP22]) is an associative, unital algebra object $L \in \mathcal{A}$ with multiplication $m : L \otimes L \rightarrow L$ and unit $\iota : \mathbb{1} \rightarrow L$ such that

- (1) (Separability) $m \circ m^* = 1$.
- (2) (Frobenius) $(1_L \otimes m^*) \circ (m \otimes 1_L) = m^* \circ m = (m \otimes 1_L) \circ (1_L \otimes m^*)$.

For a more detailed exposition on Q-systems, we refer the reader to [CHJP22] (c.f. [BKLR15]). L is *connected* if $\dim(\mathcal{A}(\mathbb{1}, L)) = 1$ and if $\mathcal{A}$ is braided with braiding σ , L is commutative if $m \circ \sigma_{L,L} = m$.

If we pick a connected, commutative Q-system $L \in \mathcal{Z}(\mathcal{C})$, consider the *category* $\mathcal{C}_L$ of *right L-modules in C* (technically we should first apply the forgetful functor to L to obtain

an object in $\mathcal{C}$). The commutative central structure on L equips $\mathcal{C}_L$ with the structure of a unitary multi-fusion category such that the free module functor $F_L : \mathcal{C} \rightarrow \mathcal{C}_L$, $x \mapsto x \otimes L$, is a dominant tensor functor (for example, see [DMNO13, Remark 3.19]).

A theorem of [BN10] shows that any quotient $F : \mathcal{C} \rightarrow \mathcal{D}$ is of the form described above, i.e. there exists a (unique up to isomorphism) connected commutative Q -system $L \in \mathcal{Z}(\mathcal{C})$ such that the diagram of monoidal functors commutes up to natural isomorphism

$$\begin{array}{ccc} \mathcal{C} & & \\ F \downarrow & \searrow F_L & \\ \mathcal{D} & \xrightarrow{\cong} & \mathcal{C}_L \end{array}$$

This gives an “internal” description of a quotient as opposed to the a-priori “external” description requiring an indecomposable multi-fusion category $\mathcal{D}$. This is directly analogous to the first isomorphism theorem for groups.

Now we recall a concrete model for constructing $\mathcal{C}_L$ and the free module functor $F_L : \mathcal{C} \rightarrow \mathcal{C}_L$. Let $L \in \mathcal{Z}(\mathcal{C})$ be a connected, commutative Q -system. We denote the multiplication morphism $m : L \otimes L \rightarrow L$ and the unit map $\iota : \mathbb{K} \rightarrow L$. We slightly abuse notation and identify L and its structure maps with their images in $\mathcal{C}$ under the forgetful functor.

Define the category $\mathcal{C}_L^\circ$ whose objects are objects of $\mathcal{C}$ and whose morphisms are given by

$$\mathcal{C}_L^\circ(x, y) := \mathcal{C}(x, y \otimes L).$$

Composition of morphisms $f \in \mathcal{C}_L^\circ(y, z)$ and $g \in \mathcal{C}_L^\circ(x, y)$ is defined as

$$f \circ g := (1_z \otimes m) \circ (f \otimes 1_L) \circ g \in \mathcal{C}_L^\circ(x, z).$$

Following [JP17], it is not hard to see that $\mathcal{C}_L^\circ$ becomes a C^* -category with $*$ -structure (denoted here with a $\dagger$ to distinguish from the $*$ -structure in $\mathcal{C}$)

$$g^\dagger := g^* \circ (1_y \otimes (m^* \circ \iota)).$$

To define the tensor structure on $\mathcal{C}_L^\circ$, the tensor product on objects (which we denote with $\boxtimes$) is just the tensor product in $\mathcal{C}$, $x \boxtimes y = x \otimes y$ while the tensor product of morphisms is defined, for $f \in \mathcal{C}_L^\circ(x, y)$ and $g \in \mathcal{C}_L^\circ(z, w)$ by

$$f \boxtimes g := (1_{y \otimes w} \otimes m) \circ (1_y \otimes \sigma_{L, w} \otimes 1_L) \circ (f \otimes g).$$

This turns $\mathcal{C}_L$ into a (strict) unitary multi-tensor category, and since L was connected in $\mathcal{Z}(\mathcal{C})$, it is in fact an indecomposable multi-fusion category. In fact, the functor $\mathcal{C}_L^\circ$ to the category FreeMod_L of free (internal to $\mathcal{C}$) right L -modules sending $x \mapsto x \otimes L$ and $f \mapsto (1_y \otimes m) \circ (f \otimes 1_L)$ is an equivalence of categories [JP17].

Thus the (unitary) Cauchy completion of $\mathcal{C}_L^\circ$ is equivalent to the category of projective right L -modules, which is equivalent to the category of all right L modules since this category is semisimple. By a minor abuse of notation, we denote the unitary Cauchy completion of $\mathcal{C}_L^\circ$ by $\mathcal{C}_L$ and identify $\mathcal{C}_L^\circ$ with its image in $\mathcal{C}_L$.

In this picture, the free module functor $F_L : \mathcal{C} \rightarrow \mathcal{C}_L$ is just given by $F_L(x) = x \in \mathcal{C}_L$ and $F_L(f) := (1_y \otimes \iota) \circ f$.

Now we have a nice way to describe the net obtained from categorical quotients $F : \mathcal{C} \rightarrow \mathcal{D}$ directly in terms of $\mathcal{C}$. Let L be the commutative Q-system in the center corresponding to this quotient, and F_L the model for F as above. Then for an interval $I \subseteq \mathbb{Z}$ with $|I| = n$,

$$A(\mathcal{C}_L, X)_I = \mathcal{C}_L(X^{\otimes n}, X^{\otimes n}) \cong \mathcal{C}(X^{\otimes n}, X^{\otimes n} \otimes L).$$

Furthermore, for $I \subseteq J$, the inclusion $A(\mathcal{C}_L, X)_I \hookrightarrow A(\mathcal{C}_L, X)_J$ is described for $f \in \mathcal{C}(X^{\otimes n}, X^{\otimes n} \otimes L)$ by

$$f \mapsto 1_{X^{\otimes l+n}} \sigma_{L, X^{\otimes r}} (1_{X^{\otimes l}} \otimes f \otimes 1_{X^{\otimes r}}).$$

Here l is the number of points in J strictly less than I , and r is the number of points in J strictly greater than I .

The inclusion $A(\mathcal{C}, X)_I \hookrightarrow A(\mathcal{C}_L, X)_I$ is simply given by

$$f \mapsto f \otimes \iota,$$

for $f \in \mathcal{C}(X^{\otimes n}, X^{\otimes n})$.

This generalizes to give a clean description of the lattice of intermediate categorical inclusions. For any sub Q-system $K \leq L$, the inclusion map $j_K : K \rightarrow L$ gives the natural maps

$$A(\mathcal{C}_K, X)_I \cong \mathcal{C}(X^{\otimes n}, X^{\otimes n} \otimes K) \hookrightarrow \mathcal{C}(X^{\otimes n}, X^{\otimes n} \otimes L) \cong A(\mathcal{C}_L, X)_I,$$

$$f \mapsto (1_{X^{\otimes n}} \otimes j_K) \circ f.$$

This sets up an equivalence between the lattice of intermediate categorical inclusions between $A(\mathcal{C}, X)$ and $A(\mathcal{C}_L, X)$ and the lattice of sub Q-system objects $K \leq L$. We typically take the largest algebra L to be normalized so that $m \circ m^* = 1_L$, but for intermediate subalgebras, we only require that $m \circ m^* = \lambda 1_K$ for some (necessarily positive) scalar λ .

Remark 4.1. Canonical conditional expectation. For any connected Q-system $L \in \mathcal{Z}(\mathcal{C})$, identifying $A(\mathcal{C}, X) \hookrightarrow A(\mathcal{C}_L, X)$, there is a canonical, locality-preserving conditional expectation $E : A(\mathcal{C}_L, X) \rightarrow A(\mathcal{C}, X)$ which is defined for $f \in A(\mathcal{C}_L, X)_I = \mathcal{C}(X^{\otimes n}, X^{\otimes n} \otimes L)$ by

$$E(f) := \frac{1}{d_L} (1_{X^{\otimes n}} \otimes i^*) \circ f.$$

Remark 4.2. By the above discussion, spatial realizations of $A(\mathcal{C}, X)$ are parameterized by choices of indecomposable $\mathcal{C}$ -module categories. Every such module category $\mathcal{M}$ canonically defines a *Lagrangian algebra* $Z(\mathcal{M}) \in \mathcal{Z}(\mathcal{C})$, its center [Dav10]. A Lagrangian algebra ([DMNO13]) in a unitary modular category (in this case, $\mathcal{Z}(\mathcal{C})$) is a connected commutative Q-system L such that the category of local modules $\mathcal{Z}(\mathcal{C})_L^{\text{loc}} \cong \text{Hilb}_{f.d.}$. In $\mathcal{Z}(\mathcal{C})$, a connected commutative Q-system is Lagrangian if and only if $d_L = \sum_{X \in \text{Irr}(\mathcal{C})} d_X^2$.

One way to identify $Z(\mathcal{M})$ is to consider the tensor functor $F : \mathcal{C} \rightarrow \text{End}(\mathcal{M})$ defined by the module category, which also defines the spatial realization at the level of spin chains. We then apply the internal description of such a quotient explained above,

which is described by an associated commutative Q-system in $\mathcal{Z}(\mathcal{C})$. However, when constructed from an indecomposable module category in this way, the commutative Q-system we obtain is a Lagrangian algebra, and every Lagrangian algebra arises in this way from a uniquely determined equivalence class of indecomposable module category.

In particular, we can always view the spatial realizations $A(\mathcal{C}, X) \hookrightarrow A(\mathcal{C}, X)_{\mathcal{M}}$ as $A(\mathcal{C}, X) \hookrightarrow A(\mathcal{C}_{\mathcal{Z}(\mathcal{M})}, X)$.

We now prove a lemma necessary for the proof of the main theorem, concerning relative commutants. The proof of this relies on the machinery of Q-system realizations for C*-algebra inclusions, developed in [CHJP22]. However, we refer the reader to Section 4.1 of [Jon24], where a concise description of the main results and definitions of the general theory are provided, with an eye specifically toward application of fusion spin chains.

As mentioned above, fusion spin chains satisfy weak algebraic Haag duality, and in fact, satisfy an even stronger version simply called algebraic Haag duality. In particular, if we choose n such that $X^{\otimes n}$ contains a copy of every simple object in $\mathcal{C}$ (the existence of such an n is the definition of strong tensor generating), then we have that for any interval I with $|I| > n$, $A(\mathcal{C}, X)_I = \{x \in A \mid [x, y] = 0 \text{ for all } y \in A(\mathcal{C}, X)_{I^c}\}$. The following lemma shows that the nets $A(\mathcal{C}_L, X)$ satisfy an even stronger “relative” version of this property.

Lemma 4.3. *Let $A(\mathcal{C}, X)$ be a fusion spin chain and n as above. For any connected, commutative Q-system $L \in \mathcal{Z}(\mathcal{C})$, then for any interval I with $|I| > n$,*

$$A(\mathcal{C}_L, X)_I = \{x \in A(\mathcal{C}_L, X) \mid [x, y] = 0 \text{ for all } y \in A(\mathcal{C}, X)_{I^c}\}.$$

Proof. We can view the quasi-local algebra $A(\mathcal{C}_L, X)$ as a finite index extension of $A(\mathcal{C}, X)_{I^c}$, supported on the canonical copy of $\mathcal{C} \boxtimes \mathcal{C}^{mp} \hookrightarrow \text{Bim}(A_{I^c})$ [Jon24, Section 4.2]. Indeed, $A(\mathcal{C}_L, X)$ is the Q-system realization of the Q-system $K_I \in \mathcal{C} \boxtimes \mathcal{C}^{mp}$ obtained from considering $\mathcal{C}_L$ as a $\mathcal{C}$ - $\mathcal{C}$ bimodule category, and choosing the object $X^{\otimes n} \in \mathcal{C}_L$, where $|I| = n$. Then set $K_I = \text{End}_{\mathcal{C} \boxtimes \mathcal{C}^{mp}}(X^{\otimes n})$, the internal endomorphism of the object $X^{\otimes n}$. Using the notation $|K_I|$ for the realization of the Q-system (in the sense of [CHJP22]), we have

$$A(\mathcal{C}_L, X) \cong |K_I|$$

Since the embedding of $\mathcal{C} \boxtimes \mathcal{C}^{mp} \hookrightarrow \text{Bim}(A(\mathcal{C}, X)_{I^c})$ is fully faithful (this follows from Ocneanu compactness, see the discussion in Sections 4.1 and 4.2 of [Jon24] for details) by the usual Q-system arguments from subfactor theory [Jon24, CHJP22].

In particular, we recall the standard subfactor theory result (translated to the C*-algebra setting by [CHJP22]) that if $A \subseteq B$ is a finite index inclusion of C*-algebras and $B \cong |P|$ is the realization of the Q-system P supported on some full subcategory $\mathcal{D} \subseteq \text{Bim}(A)$, then the relative commutant $A' \cap B$ is naturally identified with the algebra $\mathcal{D}(\mathbb{1}_{\mathcal{D}}, P)$, equipped with the convolution product that we used earlier in our “internal description” of categorical inclusions of spin chains. In our case, this means that we have the following chain of algebra isomorphisms

$$\begin{aligned} A(\mathcal{C}, X)'_{I^c} \cap A(\mathcal{C}_L, X) &\cong \mathcal{C} \boxtimes \mathcal{C}^{mp}(\mathbb{1}, K_I) \\ &= \mathcal{C} \boxtimes \mathcal{C}^{mp}(\mathbb{1}, \text{End}(X^{\otimes n})). \end{aligned}$$

By the universal property of internal end, we have the above algebra isomorphic to

$$\text{End}_{\mathcal{C}_L}(X^{\otimes n}) \cong A(\mathcal{C}_L, X)_I.$$

Thus the inclusion $A(\mathcal{C}_L, X)_I \hookrightarrow A(\mathcal{C}, X)'_{I^c} \cap A(\mathcal{C}_L, X)$ is necessarily an isomorphism, as desired. $\square$

5. Solving the Extension Problem with DHR Bimodules

Here, we recall the definition of a DHR bimodule on a net of algebras (over $\mathbb{Z}$). First, if A is a (unital) C^* -algebra, then a *bimodule* is an algebraic A - A bimodule X , together with a right A -valued inner product $\langle \cdot | \cdot \rangle : X \times X \rightarrow A$ (making X into a Hilbert A -module) such that the left A action is by adjointable operators. What we call bimodules are more typically called *correspondences* in the C^* -algebra literature.

We denote the collection of all A - A bimodules by $\text{Bim}(A)$. This forms a C^* -tensor category, whose objects are bimodules, morphisms are adjointable bimodule intertwiners, and tensor product is given by the relative tensor product of bimodules, denoted $\boxtimes_A$ (see [CHJP22]).

If $X \in \text{Bim}(A)$, a *projective basis* is a finite set $\{b_i\} \subseteq X$ such that for all $x \in X$

$$x = \sum_i b_i \langle b_i | x \rangle.$$

Definition 5.1. Let A be an abstract spin chain with quasi-local algebra A . A bimodule over the quasilocal algebra A is called a *DHR bimodule* if there exists an R such that for every interval with $|I| > R$, there exists a projective basis $\{b_i\}$ for X such that $ab_i = b_i a$ for all $a \in A_{I^c}$.

The collection of DHR bimodules is closed under $\boxtimes_A$ and thus defines a full $\otimes$ subcategory of $\text{DHR}(A) \subseteq \text{Bim}(A)$. If A also satisfies *weak algebraic Haag duality* (a property satisfied by fusion spin chains, see [Jon24]), then $\text{DHR}(\mathfrak{A})$ admits a *canonical braiding*.

The clearest physical interpretation of DHR bimodules as a braided tensor category is in terms of bulk patch operators. One realization of a fusion spin chain is as the “boundary algebra” of 2+1D topologically ordered spin system, in particular of Levin-Wen type models [LW05, JNPW23]. In this case, string operators associated to an anyon type $Z \in \mathcal{Z}(\mathcal{C})$ in the bulk can be pushed onto the boundary, giving operators in the fusion spin chain $A(\mathcal{C}, X)$ called *patch operators* [CW23, IW23]. If we push the right endpoint off to infinity, this operator no longer lives in $A(\mathcal{C}, X)$, but can be thought of formally as an element in a DHR-bimodule since it is localized in an interval around the left endpoint. Similarly to the GNS construction, we can take local perturbations and assemble this into a DHR-bimodule, corresponding to the anyon type $Z \in \mathcal{Z}(\mathcal{C})$. This picture is discussed in detail in [JNPW23].

The above discussion suggests there should be a close relationship between $\text{DHR}(A(\mathcal{C}, X))$ and $\mathcal{Z}(\mathcal{C})$, which is demonstrated in the following theorem.

Theorem 5.2 [Jon24, Theorem C]. *If X is a strong, self-dual tensor generator of the unitary fusion category $\mathcal{C}$, then $\text{DHR}(A(\mathcal{C}, X)) \cong \mathcal{Z}(\mathcal{C})$ as braided C^* -tensor categories.*

We note that more recent results [HJJY25, JNP25] allow us to remove the self-duality assumption, but this is rarely (if ever) considered in practice.

The above theorem provides an MPO picture for DHR bimodules, in the case of a spatial realization. First, for an object $Z \in \mathcal{Z}(\mathcal{C})$, we consider defect tensors a as in the

following picture, where the horizontal string is labeled by an object Z in the Drinfeld center, and M is the associated MPO for the object $Z \in \mathcal{C}$.

Diagram (5.3) shows a horizontal string with a box labeled a followed by three boxes labeled M . The string is represented by two vertical lines on the left and right, with ellipses indicating continuation. The equation is labeled (5.3).

See [BMW+17, WBV17] for a description of how to decompose the space of defect tensors for MPOs corresponding to a category $\mathcal{C}$ into simple objects of the center $\mathcal{Z}(\mathcal{C})$.

Next, we can represent the half-braiding associated to Z with an MPO h , and in this way extend the action of the fusion ring by channels on local operators to the defect tensors via the following picture.

Diagram (5.4) shows a horizontal string with a box labeled a and a box labeled h connected by a loop. The string is represented by two vertical lines on the left and right, with ellipses indicating continuation. The equation is labeled (5.4).

The defect sectors are decomposed into irreducible representations under this action, which is similar to Ocneanu's tube algebra [Ocn88].

Finally, we consider the vector space of all such defect tensors invariant under the $\mathcal{C}$ -symmetry. This is a bimodule under the action of symmetric local operators $L_1, L_2, \dots$ and $R_1, R_2, \dots$ as depicted in the following picture.

Diagram (5.5) shows a horizontal string with a box labeled a and a box labeled M . The string is surrounded by boxes labeled $L_1, L_2, L_3, L_4, R_1, R_2, R_3$. The string is represented by two vertical lines on the left and right, with ellipses indicating continuation. The equation is labeled (5.5).

By the above theorem, it turns out that these constitute *all* the abstractly defined DHR bimodules in the symmetric case.

One of the main motivations of [Jon24] is that the DHR construction defines a functor $\text{DHR} : \text{Net}_{\mathbb{Z}} \rightarrow \text{C}^*\text{-BrTens}$, where $\text{Net}_{\mathbb{Z}}$ is the groupoid of abstract spin chains and

bounded-spread isomorphisms. This associates to each abstract spin chain its braided tensor category of DHR bimodules, while

$$\mathrm{DHR}(\alpha) : \mathrm{DHR}(A) \rightarrow \mathrm{DHR}(B)$$

which acts on DHR bimodules by sending a DHR bimodule X of A to $\mathrm{DHR}(\alpha)(X) := X$ as a vector space, with left and right B actions given by

$$b \triangleright x \triangleleft c := \alpha^{-1}(b)x\alpha^{-1}(c),$$

with B valued inner product

$$\langle x \mid y \rangle_B := \alpha(\langle x \mid y \rangle_A).$$

In particular, for every bounded-spread isomorphism between fusion spin chains

$$\alpha : A(\mathcal{C}, X) \rightarrow A(\mathcal{D}, Y),$$

there is an induced unitary braided equivalence of categories

$$\mathrm{DHR}(\alpha) : \mathcal{Z}(\mathcal{C}) \cong \mathrm{DHR}(A(\mathcal{C}, X)) \rightarrow A(\mathcal{D}, Y) \cong \mathcal{Z}(\mathcal{D}).$$

Theorem 5.6. *Let $\alpha : A(\mathcal{C}, X) \rightarrow A(\mathcal{D}, Y)$ be a bounded-spread isomorphism. Then for any commutative, connected $\mathcal{Q}$ -systems $L \in \mathcal{Z}(\mathcal{C})$ and $K \in \mathcal{Z}(\mathcal{D})$, extensions of α to bounded-spread isomorphisms $\tilde{\alpha} : A(\mathcal{C}_L, X) \rightarrow A(\mathcal{D}_K, Y)$ are in bijective correspondence with algebra isomorphisms $\gamma : \mathrm{DHR}(\alpha)(L) \cong K$ in $\mathcal{Z}(\mathcal{D})$. As a consequence:*

- (1) *If $\mathrm{DHR}(\alpha)(L)$ is not isomorphic to K , then α has no extension to a bounded-spread isomorphism $\tilde{\alpha} : A(\mathcal{C}_L, X) \rightarrow A(\mathcal{D}_K, Y)$*
- (2) *If $\mathrm{DHR}(\alpha)(L) \cong K$, bounded spread extensions form a torsor over $\mathrm{Aut}(L)$.*

Proof. Let $L \in \mathcal{Z}(\mathcal{C})$ be a commutative, connected $\mathcal{Q}$ -system. Then $A(\mathcal{C}_L, X)$ is algebraically a bimodule over the quasi-local algebra $A(\mathcal{C}, X)$ obtained by including the latter as a subalgebra of the former. The canonical condition expectation equips $A(\mathcal{C}_L, X)$ with the structure of a right Hilbert $A(\mathcal{C}, X)$ module, hence $(A(\mathcal{C}_L, X), E) \in \mathrm{Bim}(A(\mathcal{C}, X))$. In fact, from the canonical embedding $\mathcal{Z}(\mathcal{C}) \cong \mathrm{DHR}(A(\mathcal{C}, X)) \subseteq \mathrm{Bim}(A(\mathcal{C}, X))$, we have that the quasi-local algebra with condition expectation $(A(\mathcal{C}_L, X), E)$ is simply the $\mathcal{Q}$ -system realization $|L|$ of $L \in \mathcal{Z}(\mathcal{C})$, and in particular $(A(\mathcal{C}_L, X), E) \in \mathrm{DHR}(A(\mathcal{C}, X))$ is a connected, commutative $\mathcal{Q}$ -system.

With this picture in hand, to establish our bijection, we simply show that the data of one thing can simply be reinterpreted as the data of the other.

First, suppose $\tilde{\alpha} : A(\mathcal{C}_L, X) \rightarrow A(\mathcal{D}_K, Y)$ is a bounded-spread isomorphism, restricting to α on the subsystem $A(\mathcal{C}, X)$. We claim that $\tilde{\alpha}$ is an algebra object isomorphism $\mathrm{DHR}(\alpha)(A(\mathcal{C}_L, X)) \cong A(\mathcal{D}_K, X)$. Indeed, $\mathrm{DHR}(\alpha)(A(\mathcal{C}_L, X))$ is simply $A(\mathcal{C}_L, X)$ with left and right action twisted by α^{-1} . We check for $a \in A(\mathcal{C}_L, X)$ and $b, c \in A(\mathcal{D}, Y)$

$$\tilde{\alpha}(b \triangleright a \triangleleft c) = \tilde{\alpha}(\alpha^{-1}(b)a\alpha^{-1}(c)) = b\tilde{\alpha}(a)c$$

where we have used $\tilde{\alpha}|_{A(\mathcal{C}, X)} = \alpha$. Thus $\tilde{\alpha} : \mathrm{DHR}(\alpha)(A(\mathcal{C}_L, X)) \cong A(\mathcal{D}_K, X)$ is an isomorphism of $A(\mathcal{D}, Y)$ bimodules. Since it is an algebra isomorphism, this bimodule isomorphism is an isomorphism of algebra objects.

Conversely, suppose we have an isomorphism of algebra objects $\tilde{\alpha} : \text{DHR}(\alpha)(A(\mathcal{C}_L, X)) \cong A(\mathcal{D}_K, X)$ in the category of $A(\mathcal{D}, Y)$ bimodules. Then reversing the above logic, $\tilde{\alpha}$ is an algebra isomorphism $A(\mathcal{C}_L, X) \rightarrow A(\mathcal{D}_K, Y)$ which restricts to α on the algebra $A(\mathcal{C}, X)$. It remains to show that $\tilde{\alpha}$ is bounded spread.

Suppose α has spread at most R . We claim $\tilde{\alpha}$ has spread at most R . Indeed, let $a \in A(\mathcal{C}_L, X)_I$. Note that $A(\mathcal{D}, Y)_{(I+R)^c} \subseteq \alpha(A(\mathcal{C}, X)_{I^c})$, and since $[a, b] = 0$ for all $b \in A(\mathcal{C}, X)_{I^c}$,

$$0 = [\tilde{\alpha}(a), \tilde{\alpha}(b)] = [\tilde{\alpha}(a), \alpha(b)],$$

and hence

$$[\tilde{\alpha}(A(\mathcal{C}_L, X)_I), A(\mathcal{D}, Y)_{(I+R)^c}] = 0.$$

By Lemma 4.3, this implies

$$\tilde{\alpha}(A(\mathcal{C}_L, X)_I) \subseteq A(\mathcal{D}_K, Y)_{I+R},$$

so that $\tilde{\alpha}$ is bounded spread. This concludes the proof of the main statement of the theorem.

The first consequence listed above follows immediately. The second follows from the fact that for any isomorphisms $\gamma, \gamma' : \text{DHR}(\alpha)(L) \cong K$, $\gamma^{-1} \circ \gamma' \in \text{Aut}(\text{DHR}(\alpha)(L) \cong \text{Aut}(L))$. $\square$

We recommend the reader to compare the above theorem with an analogous result concerning extending symmetries in 1+1D algebraic quantum field theory [BJLP19, Section 6].

Proof of Theorem 1.3. The Theorem 1.3 stated in the introduction follows as a special case of the above, and we believe it provides a satisfactory answer to Question 2. Indeed, consider the case where $L \in \mathcal{Z}(\mathcal{C})$ and $K \in \mathcal{Z}(\mathcal{D})$ are $Z(\mathcal{M})$ and $Z(\mathcal{N})$ associated to the $\mathcal{C}$ and $\mathcal{D}$ module categories, respectively. Then we have identifications of the spatial realizations $A(\mathcal{C}, X)_{\mathcal{M}} \cong A(\mathcal{C}_{Z(\mathcal{M})}, X)$, whence the first part of the result follows from the previous theorem. That $\text{Aut}(Z(\mathcal{M})) \cong \text{Inv}(\mathcal{C}_{\mathcal{M}}^*)$ follows, for example, from [BJ22]. $\square$

For any algebra $L \in \mathcal{Z}(\mathcal{C})$, there is a group

$$\text{Aut}_{br}(\mathcal{Z}(\mathcal{C}) \mid L)$$

which consists of equivalence classes of pairs (α, γ) , where α is a braided tensor equivalence of $\mathcal{Z}(\mathcal{C})$ and $\gamma : \alpha(L) \cong L$ is an isomorphism of algebras. An equivalence between pairs (α, γ) and (β, δ) consists of a monoidal natural isomorphism $v : \alpha \cong \beta$ such that $\delta \circ v_L = \gamma$. The composition is defined as $(\alpha, \gamma) \circ (\beta, \delta) := (\alpha \circ \beta, \gamma \alpha(\delta))$, which is well-defined on equivalence classes. By [DMNO13, Sch24], for a Lagrangian algebra L corresponding to an indecomposable $\mathcal{C}$ -module category $\mathcal{M}$, we have

$$\text{Aut}_{br}(\mathcal{Z}(\mathcal{C}) \mid L) \cong \text{Aut}_{\otimes}(\mathcal{C}_{\mathcal{M}}^*)$$

where the latter is the group of tensor autoequivalences up to monoidal natural isomorphism.

Let $\text{QCA}(\mathcal{C}, X, \mathcal{M})$ be the group of bounded-spread autoequivalences of $A(\mathcal{C}, X)_{\mathcal{M}}$ which preserve the subalgebra $A(\mathcal{C}, X)$. We can think of this as the group of spatially implemented dualities, or as the group of “edge-restricted” QCA that preserve the symmetric subalgebra. Then the above theorem yields the following corollary:

Corollary 5.7. *There is a homomorphism $\pi : \text{QCA}(\mathcal{C}, X, \mathcal{M}) \rightarrow \text{Aut}_{\otimes}(\mathcal{C}_{\mathcal{M}}^*)$.*

Proof. By the above theorem for any $\tilde{\alpha} \in \text{QCA}(\mathcal{C}, X, \mathcal{M})$, $\alpha := \tilde{\alpha}|_{A(\mathcal{C}, X)}$ is a bounded spread isomorphism, and the choice $\tilde{\alpha}$ of extension specifies an algebra isomorphism $\gamma : \text{DHR}(\alpha)(Z(\mathcal{M})) \cong Z(\mathcal{M})$. Thus for any $\tilde{\alpha}$, we have an element $(\tilde{\alpha}, \gamma) \in \text{Aut}_{br}(\mathcal{Z}(\mathcal{C})|Z(\mathcal{M})) \cong \text{Aut}_{\otimes}(\mathcal{C}_{\mathcal{M}}^*)$ gives the desired result π . It is easy to check that this is a homomorphism. $\square$

We can easily choose examples so that this homomorphism is surjective. Indeed, consider the object $X = \bigoplus_{Y \in \text{Irr}(\mathcal{C})} Y$. Then $\pi : \text{QCA}(\mathcal{C}, X, \mathcal{C}) \rightarrow \text{Aut}_{\otimes}(\mathcal{C})$ is surjective (see [Jon24, Theorem 4.25]).

A family of examples of interest, which arise from onsite finite group symmetry, are when $\mathcal{C} = \text{Rep}(G)$, $\mathcal{M} = \text{Hilb}_{f.d.}$, made into a $\mathcal{C}$ -module category with the standard fiber functor structure on $\text{Rep}(G)$. In this case, we obtain a homomorphism

$$\pi : \text{QCA}(\text{Rep}(G), X, \text{Hilb}_{f.d.}) \rightarrow \text{Aut}_{\otimes}(\text{Hilb}_{f.d.}(G)) \cong H^2(G, U(1)) \rtimes \text{Aut}(G).$$

6. Examples and Applications

In this section, we consider a family of examples generalizing Kramers-Wannier (KW) dualities [LDOV23], and we also discuss applications of our previous result.

Generalized shifts. Let $\mathcal{C}$ be a unitary fusion category, and $\alpha \in \text{Aut}_{br}(\mathcal{Z}(\mathcal{C}))$, such that α generates a categorical action of $\mathbb{Z}/n\mathbb{Z}$ on $\mathcal{Z}(\mathcal{C})$ for some n . Then the o_4 obstruction necessarily vanishes as $H^4(\mathbb{Z}/n\mathbb{Z}, U(1))$ is trivial, and thus by [ENO10] there exists a $\mathbb{Z}/n\mathbb{Z}$ -graded extension

$$\mathcal{D} := \bigoplus_{g \in \mathbb{Z}/n\mathbb{Z}} \mathcal{C}_g,$$

with $\mathcal{C}_e = \mathcal{C}$. For any $Y \in \mathcal{C}_\alpha$, $Y^{\otimes n} \in \mathcal{C}_e = \mathcal{C}$. We can always choose Y such that $X := Y^{\otimes n}$ is a strong tensor generator for $\mathcal{C}$, for example, $Y = \bigoplus_{Z \in \text{Irr}(\mathcal{C}_\alpha)} Z$.

Now, consider the fusion spin chain $A(\mathcal{C}, X)$. We define a “generalized translation” QCA for $a \in A(\mathcal{C}, X)_{[i, j]} \cong \mathcal{C}(X^{j-i+1}, X^{j-i+1}) \cong \mathcal{D}(Y^{n(j-i+1)}, Y^{n(j-i+1)})$,

$$\tau_\alpha(a) := 1_{Y^{n-1}} \otimes a \otimes 1_Y \in A(\mathcal{C}, X)_{[i-1, j]}.$$

Then since for any $Z \in \mathcal{Z}(\mathcal{C})$ we have natural isomorphisms $Y \otimes Z \cong \alpha(Z) \otimes Y$ (see [ENO10] or more explicitly, [GNN09, Section 3A]), it is easy to verify that

$$\text{DHR}(\tau_\alpha) \cong \alpha.$$

Thus, any braided autoequivalence of $\mathcal{Z}(\mathcal{C})$ can be implemented by a QCA on some fusion spin chain of the form $A(\mathcal{C}, X)$, although the choice of X depends on the autoequivalence.

Dualities from Tambara-Yamagami. The easiest example is the case $\mathcal{C} := \text{Hilb}_{f.d.}(A)$, where A is a finite abelian group. $\mathcal{Z}(\mathcal{C}) \cong \text{Hilb}_{f.d.}(A \times \hat{A})$. Any symmetric, non-degenerate bicharacter $\chi : A \times A \rightarrow U(1)$, yields an isomorphism $\tilde{\chi} : A \rightarrow \hat{A}$, given by $\tilde{\chi}(x) = \chi(a, \cdot)$. This results in a braided autoequivalence $\alpha_\chi \in \text{Aut}_{br}(\text{Hilb}_{f.d.}(A \times \hat{A}))$ defined by

$$\alpha_\chi(a, f) = (\tilde{\chi}^{-1}(f), \tilde{\chi}(a)).$$

Which further results in a $\mathbb{Z}/2\mathbb{Z}$ extension of $\text{Hilb}_{f.d.}(A)$ called a *Tambara-Yamagami* category [TY98], whose simple objects are simply $A \cup \{\rho\}$ with fusion rules

$$a \otimes b = ab,$$

$$a \otimes \rho = \rho = \rho \otimes a,$$

$$\rho \otimes \rho \cong \bigoplus_{a \in A} a.$$

If we set $Y = \rho$, then $X := Y^2 = \bigoplus_{a \in A} a$ is a strong tensor generator for $\mathcal{C}$, and thus the above construction yields a QCA on the spin chain

$$\tau_{\alpha_\chi} : A(\mathcal{C}, X) \rightarrow A(\mathcal{C}, X)$$

with $\text{DHR}(\tau_{\alpha_\chi}) \cong \alpha_\chi$. For $A = \mathbb{Z}_2$, there is a unique symmetric, non-degenerate bicharacter χ , and τ_{α_χ} is exactly the Kramers-Wannier duality map described earlier.

There are two canonical spatial realizations of $A(\mathcal{C}, X)$ described by the Lagrangian algebras $L_1 := \mathbb{C}[A] \times 1$ and $L_2 := 1 \times \mathbb{C}[\widehat{A}]$, sometimes called electric and magnetic algebras. Then α_χ swaps L_1 and L_2 . In particular, by Theorem 1.3, τ_{α_χ} cannot extend to a QCA on either spatial realization $A(\mathcal{C}_{L_1}, X)$ or $A(\mathcal{C}_{L_2}, X)$.

Q-system complete fusion categories. A unitary fusion category $\mathcal{C}$ is called *Q-system complete* (also called *torsion-free in the literature* if there is exactly one indecomposable module category (namely, $\mathcal{C}$ itself) up to equivalence. This terminology comes from the fact that the fusion category $\mathcal{C}$, thought of as a 2-category with one object, is Q-system complete in the sense of [CHJP22]. Examples include Fib and more generally $\text{PSU}(2)_{2k+1}$, and $\text{Hilb}_{f.d.}(\mathbb{Z}/p\mathbb{Z}, \omega)$, where $[\omega] \in H^3(\mathbb{Z}/p\mathbb{Z})$ is non-trivial.

For Q-system complete fusion categories $\mathcal{C}$, any fusion spin chain of the form $A(\mathcal{C}, X)$ has only one spatial realization $A(\mathcal{C}, X)_{\mathcal{C}}$, which in turn corresponds to a unique isomorphism class of Lagrangian algebra in $\mathcal{Z}(\mathcal{C})$. This results in the following corollary of Theorem 1.3.

Proposition 6.1. *Let $\mathcal{C}$ be a Q-system complete fusion category. Suppose $\alpha : A(\mathcal{C}, X) \rightarrow A(\mathcal{D}, Y)$ is a bounded-spread isomorphism between fusion spin chains. Then*

- (1) $\mathcal{C} \cong \mathcal{D}$ as fusion categories and in particular $\mathcal{D}$ is Q-system complete.
- (2) There exists a spatial implementation $\tilde{\alpha} : A(\mathcal{C}, X)_{\mathcal{C}} \rightarrow A(\mathcal{D}, Y)_{\mathcal{D}}$ of α .
- (3) The spatial implementations form a torsor over $\text{Inv}(\mathcal{C})$. In particular, if $\mathcal{C}$ has no non-trivial invertible objects (e.g. $\text{PSU}(2)_{2k+1}$), then there is a unique spatial implementation.

Proof. Let L denote the unique Lagrangian algebra object in $\text{DHR}(A(\mathcal{C}, X)) \cong \mathcal{Z}(\mathcal{C})$, so that $\mathcal{C} \cong \mathcal{Z}(\mathcal{C})_L$. Then $\text{DHR}(\alpha)(L) \in \text{DHR}(A(\mathcal{D}, Y)) \cong \mathcal{Z}(\mathcal{D})$ is the unique Lagrangian algebra (since $\text{DHR}(\alpha)$ is a braided equivalence). In particular, we have $K := \text{DHR}(\alpha)(L)$ must be the canonical Lagrangian algebra in $\mathcal{Z}(\mathcal{D})$, hence $\mathcal{Z}(\mathcal{D})_K \cong \mathcal{D}$, and thus $\text{DHR}(\alpha)$ induces a monoidal equivalence $\mathcal{C} \cong \mathcal{Z}(\mathcal{C})_L \rightarrow \mathcal{Z}(\mathcal{D})_K \cong \mathcal{D}$.

Since the spatial realizations must correspond to the Lagrangian algebras L and K respectively, the existence of a spatial implementation of α , and the statement about the torsor, follows immediately from Theorem 1.3. $\square$

Classification of QCA. In this section, we apply our above analysis to give a better understanding of QCA which commute with a group symmetry, and connect our perspective to the literature. Suppose we have a group G acting unitarily on the local Hilbert space $V := \mathbb{C}^d$, given by an assignment $g \mapsto U_g \in U(d)$. Let A denote the spin system. The group action extends to the automorphism of each A_I by $\alpha_g(a) := (U_g)^{\otimes n} a (U_g^*)^{\otimes n}$, which extends to a homomorphism $G \rightarrow \text{Aut}(A)$. We assume this homomorphism is injective.

In the literature, there have been several investigations into *symmetric QCA* (*sQCA*) [GSSC20], which in our language are bounded-spread isomorphisms $\alpha : A \rightarrow A$ such that $\alpha \circ \alpha_g = \alpha_g \circ \alpha$ for all $g \in G$. Our goal in this section is to make contact with these results from our perspective.

First, we recast this picture into the language of fusion spin chains by noting that A^G has the structure of an abstract spin chain with local algebras $A_I^G := \{x \in A_I \mid \alpha_g(x) = x \text{ for all } g \in G\}$.

Now if we view V as an object in $\text{Rep}(G)$, and we further assume that V is self-dual and strongly tensor generating (e.g. $V \cong L^2(G)$ is the left regular representation), then

$$A^G \cong A(\text{Rep}(G), V).$$

The inclusion $A^G \subseteq A$ is a spatial realization which corresponds to the standard fiber functor $\text{Rep}(G) \rightarrow \text{Hilb}_{f.d.}$. We remark that the actual net of algebras obtained is very sensitive to the choice of V (hence α). See [CHK+24] for details. This leads to the following natural question:

Question 6.2. *Suppose $\alpha : A^G \rightarrow A^G$ is a duality and $\tilde{\alpha} : A \rightarrow A$ is a spatial implementation. Under what conditions is the spatial implementation $\tilde{\alpha}$ a symmetric QCA?*

To answer this question, note that the Lagrangian algebra resulting from this spatial realization in $\mathcal{Z}(\text{Rep}(G))$ can actually be viewed as an algebra object in $\text{Rep}(G) \subseteq \mathcal{Z}(\text{Rep}(G))$, where we identify $\text{Rep}(G)$ with the full subcategory of its center using the standard symmetric braiding. The algebra is simply $L := \text{Fun}(G)$, the commutative algebra of $\mathbb{C}$ -valued functions, viewed as a G -representation where G acts by translation [EGNO15].

Theorem 6.3. *Suppose $\alpha : A^G \rightarrow A^G$ is a duality and $\tilde{\alpha} : A \rightarrow A$ is a spatial implementation, and $\gamma : \text{DHR}(\alpha)(L) \cong L$ is the associated algebra isomorphism. Then*

- (1) $\tilde{\alpha}$ is a symmetric QCA if and only if for every $\beta \in \text{Aut}(L)$, $\gamma \circ \text{DHR}(\alpha)(\beta) = \beta \circ \gamma$.
- (2) There is a homomorphism from the group of sQCA on A to $H^2(G, U(1))$.

Proof. A , as an algebra object (Q-system) in $\text{Bim}(A^G)$, is isomorphic to the Q-system realization $|L|$ of the function algebra $\text{Fun}(G) \in \text{Rep}(G) \subseteq \mathcal{Z}(\text{Rep}(G))$, and thus we have that the group $\text{Aut}(A|A^G)$ is naturally isomorphic to the group $\text{Aut}_{\text{Rep}(G)}(L) = G$. In particular, we see that the group elements are the extensions of the identity map on A^G , given by the automorphism of L labeled by $g \in G$. But if $\tilde{\alpha}$ is an sQCA, then it must commute with each of the automorphisms from G . Using the definition of how the extension works in the proof of Theorem 5.6, which we get (1). Part (2) follows from [Sch24, Corollary 3.1]. To elaborate, recall $\text{Aut}(\mathcal{Z}(\text{Rep}(G))|L) \cong \text{Aut}_{\otimes}(\text{Hilb}_{f.d.}(G)) \cong \text{Aut}(G) \rtimes H^2(G, U(1))$. In [Sch24, Corollary 3.1], it is shown that the elements that commute with all the automorphisms of L is precisely the subgroup $H^2(G, U(1))$. $\square$

Our next goal is to prove Corollary 1.4 from the introduction. This requires us to review the index for categorical dualities of fusion spin chains, introduced in [JL24]. Given an inclusion of C^* -algebras $A \subseteq B$ with unique tracial state τ , we can build an associated inclusion of von Neumann II_1 factors $N = A'' \subseteq M = B''$, where the completions are taken in the Hilbert space representation $L^2(A, \tau)$. Then the Jones index $[M : N]$ is a real number that satisfies several nice properties [Jon83]. Since A, B are assumed to have a unique trace, their II_1 factor completion is unambiguous, and thus we can slightly abuse notation and write $[B : A] := [M : N]$.

Let α be a bounded spread automorphism $\alpha \in \text{Aut}(A)$. For any $z \in \mathbb{Z}$, define $A_z \subseteq A$ as the C^* -algebra of observables localized in the interval $(-\infty, z]$. Fusion spin chains $A := A(\mathcal{C}, X)$ satisfy the unique trace property, as do all the algebras A_z .

Now choose $x, y \in \mathbb{Z}$ so that $y \leq x$, $A_y \subseteq \alpha(A_x)$, which can be accomplished due to the bounded spread condition.

Now we can define the index

$$\text{Ind}(\alpha) := \left(\frac{[\alpha(A_x) : A_y]}{[A_x : A_y]} \right)^{\frac{1}{2}}.$$

For ordinary spin chains, this agrees with the GNVW index [JL24, Proposition 4.2]. Our goal is to show that for the group case, if a G -duality α admits an extension to a QCA $\tilde{\alpha}$, then $\text{Ind}(\alpha) = \text{Ind}(\tilde{\alpha})$. This would imply that if $\text{Ind}(\alpha) = 1$, $\tilde{\alpha}$ is a finite depth circuit.

Proposition 6.4. *Let $\tilde{\alpha} : A \rightarrow A$ be a QCA such that $\tilde{\alpha}(A^G) = A^G$, and let $\alpha := \tilde{\alpha}|_{A^G}$. Then $\text{Ind}(\tilde{\alpha}) = \text{Ind}(\alpha)$.*

Proof. Let $y \leq x$ be as in the definition of index. Since the G -representation is a strong tensor generator, it follows that $A_x, A_y, A_x^G, A_y^G, \tilde{\alpha}(A_x), \alpha(A_x^G)$ are simple C^* -algebras with unique trace. Then we have the inclusions

$$\begin{array}{ccc} A_x^G & \subset & A_x \\ \cup & & \cup \\ A_y^G & \subset & A_y \end{array} \quad \text{and} \quad \begin{array}{ccc} \alpha(A_x^G) & \subset & \tilde{\alpha}(A_x) \\ \cup & & \cup \\ A_y^G & \subset & A_y \end{array}$$

Since the indices are actually indices of subfactors, the index is multiplicative [Jon83], hence

$$[A_x : A_x^G][A_x^G : A_y^G] = [A_x : A_y][A_y : A_y^G]$$

and

$$[\tilde{\alpha}(A_x) : \alpha(A_x^G)][\alpha(A_x^G) : A_y^G] = [\tilde{\alpha}(A_x) : A_y][A_y : A_y^G].$$

But from standard results in subfactor theory (for example [JS97, EK98]),

$$[A_x : A_x^G] = [A_y : A_y^G] = [\tilde{\alpha}(A_x) : \alpha(A_x^G)] = [A_y : A_y^G] = |G|.$$

Thus we have $[A_x^G : A_y^G] = [A_x : A_y]$ and $[\alpha(A_x^G) : A_y^G] = [\tilde{\alpha}(A_x) : A_y]$. Thus $\text{Ind}_G(\alpha) = \text{Ind}(\alpha)$. $\square$

Proof of Corollary 1.4. Suppose $\alpha : A^G \rightarrow A^G$ is a G -duality such that $\text{Ind}(\alpha) = 1$ and $\text{DHR}(\alpha) = \text{Id}_{\mathcal{Z}(\text{Rep}(G))}$. Then by Theorem 1.3 there is a lift of α to an sQCA $\tilde{\alpha}$ which can be chosen by 6.3 utilizing the identity automorphism of $L = \text{Fun}(G)$ to have trivial H^2 index. Furthermore, by Proposition 6.4, $\text{Ind}(\tilde{\alpha}) = 1$, so $\tilde{\alpha}$ is a finite depth circuit.

Now, for abelian G in the regular representation, it was recently shown that globally symmetric circuits with trivial H^2 invariant are locally symmetric [MLC24, Theorem 1]. More generally, the corresponding statement for groups G , which are not necessarily abelian, is shown (implicitly¹) in [LE24]. This concludes the proof. $\square$

Acknowledgements. CJ and KS were supported by NSF DMS-2247202. DJW was supported in part by the Australian Research Council Discovery Early Career Research Award (DE220100625). Part of this work was done while DJW was working at IBM Quantum, IBM Almaden Research Center, San Jose, CA 95120, USA. This paper grew out of discussions that took place during the “QCA group” at the workshop “Fusion categories and tensor networks” at the American Institute of Mathematics in 2022. After completing the first arXiv version of this work we became aware of the related work [MLC24]. We have now included comments on the relationship between our results and those of [MLC24]. We also thank Dominic Else for pointing out how the results of [LE24] can be used to obtain Corollary 1.4, which is a stronger statement than in earlier versions.

Funding Open Access funding enabled and organized by CAUL and its Member Institutions.

Data Availability No data was generated during this work.

Declarations

Conflict of interest The authors declare no conflict of interest.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article’s Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article’s Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

Publisher’s Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

References

- [AFM20] Aasen, D., Fendley, P., Mong, R.: Topological defects on the lattice: dualities and degeneracies (2020)
- [AHLM23] Aasen, D., Haah, J., Li, Z., Mong, R.: Measurement quantum cellular automata and anomalies in floquet codes (2023)
- [AMF16] Aasen, D., Mong, R., Fendley, P.: Topological defects on the lattice: I. the Ising model. *J. Phys. A Math. Theor.* **49**(35), 354001 (2016)
- [ANW11] Arrighi, P., Nesme, V., Werner, R.: Unitarity plus causality implies localizability. *J. Comput. Syst. Sci.* **77**(2), 372–378 (2011). Adaptivity in Heterogeneous Environments
- [Arr19] Arrighi, P.: An overview of quantum cellular automata (2019)
- [BDSNY24] Bhardwaj, L., Décoppet, T., Schafer-Nameki, S., Yu, M.: Fusion 3-Categories for Duality Defects. arXiv preprint (2024)
- [BJ22] Bischoff, M., Jones, C.: Computing fusion rules for spherical G -extensions of fusion categories. *Sel. Math. (N.S.)* **28**(2), 26, 39 (2022)

¹ The argument stated there uses ancilla but can easily be modified to avoid this step by coarse-graining and applying Fell’s absorption principle in the regular representation.

- [BJLP19] Bischoff, M., Jones, C., Yuan Ming, L., Penneys, D.: Spontaneous symmetry breaking from anyon condensation. *J. High Energy Phys.* **2019**, 11 (2019)
- [BKLR15] Bischoff, M., Kawahigashi, Y., Longo, R., Rehren, K.-H.: Tensor categories and endomorphisms of von Neumann algebras—with applications to quantum field theory. *SpringerBriefs in Mathematical Physics*, vol. 3. Springer, Cham (2015)
- [BM20] Brun, T.A., Młodinow, L.: Quantum cellular automata and quantum field theory in two spatial dimensions. *Phys. Rev. A* **102**(6) (2020)
- [BMW+17] Bultinck, N., Mariën, M., Williamson, D., Şahinoğlu, M.B., Haegeman, J., Verstraete, F.: Anyons and matrix product operator algebras. *Ann. Phys.* **378**, 183–233 (2017)
- [BN10] Bruguières, A., Natale, S.: Exact sequences of tensor categories. *Int. Math. Res. Not.* **5644–5705**(6), 2010 (2011)
- [BR87] Bratteli, O., Robinson, D.: *Operator Algebras and Quantum Statistical Mechanics. 1. Texts and Monographs in Physics*, 2nd edn. Springer, New York (1987). C^* - and W^* -algebras, symmetry groups, decomposition of states
- [BR97] Bratteli, O., Robinson, D.: *Operator Algebras and Quantum Statistical Mechanics. 2. Texts and Monographs in Physics*, 2nd edn. Springer, Berlin (1997). Equilibrium states. Models in quantum statistical mechanics
- [CCH+21] Choi, Y., Córdova, C., Hsin, P.S., Lam, H.T., Shao, S.H.: Non-invertible duality defects in 3+1 dimensions. *Phys. Rev. D* **105**(12) (2021)
- [CGW10] Chen, X., Zheng-Cheng, G., Wen, X.-G.: Local unitary transformation, long-range quantum entanglement, wave function renormalization, and topological order. *Phys. Rev. B* **82**, 155138 (2010)
- [CHK+24] Chuah, C.Y., Hungar, B., Kawagoe, K., Penneys, D., Tomba, M., Wallick, D., Wei, S.: Boundary algebras of the Kitaev quantum double model. *J. Math. Phys.* **65**(10), 102201 (2024)
- [CHJP22] Chen, Q., Palomares, R.H., Jones, C., Penneys, D.: Q-system completion for C^* 2-categories. *J. Funct. Anal.* **283**(3), 109524 (2022)
- [CPGSV17] Cirac, J.I., Perez-Garcia, D., Schuch, N., Verstraete, F.: Matrix product unitaries: structure, symmetries, and topological invariants. *J. Stat. Mech. Theory Exp.* **2017**(8), 083105 (2017)
- [CW23] Chatterjee, A., Wen, X.-G.: Symmetry as a shadow of topological order and a derivation of topological holographic principle. *Phys. Rev. B* **107**, 155136 (2023)
- [Dav10] Davydov, A.: Centre of an algebra. *Adv. Math.* **225**(1), 319–348 (2010)
- [DMNO13] Davydov, A., Müger, M., Nikshych, D., Ostrik, V.: The Witt group of non-degenerate braided fusion categories. *J. Reine Angew. Math.* **135–177**, 9 (2013)
- [ECN11] Emilio Cobanera, G.O., Nussinov, Z.: The bond-algebraic approach to dualities. *Adv. Phys.* **60**(5), 679–798 (2011)
- [EGNO15] Etingof, P., Gelaki, S., Nikshych, D., Ostrik, V.: *Tensor Categories. Mathematical Surveys and Monographs*, vol. 205. American Mathematical Society, Providence (2015)
- [EK98] Evans, D., Kawahigashi, Y.: *Quantum Symmetries on Operator Algebras. Oxford Mathematical Monographs. The Clarendon Press, Oxford University Press, New York* (1998). Oxford Science Publications
- [ENO10] Etingof, P., Nikshych, D., Ostrik, V.: Fusion categories and homotopy theory. *Quantum Topol.* **1**(3), 209–273 (2010) With an appendix by Ehud Meir
- [Far20] Farrelly, T.: A review of quantum cellular automata. *Quantum* **4**, 368 (2020)
- [FFRS04] Fröhlich, J., Fuchs, J., Runkel, I., Schweigert, C.: Kramers-Wannier duality from conformal defects. *Phys. Rev. Lett.* **93**(7), 70601 (2004)
- [FH20] Freedman, M., Hastings, M.: Classification of quantum cellular automata. *Commun. Math. Phys.* **376**, 06 (2020)
- [FHH19] Freedman, M., Haah, J., Hastings, M.: The group structure of quantum cellular automata. *Commun. Math. Phys.* **389**, 1277–1302 (2019)
- [FHH20] Fidkowski, L., Haah, J., Hastings, M.B.: Exactly solvable model for a 4+1D beyond-cohomology symmetry-protected topological phase. *Phys. Rev. B* **101**, 155124 (2020)
- [FHP20] Ferrer, G., Palomares, R.H.: Classifying module categories for generalized Temperley-Lieb-Jones $*2$ -categories. *Int. J. Math.* **31**(04), 2050027 (2020)
- [FS20] Farrelly, T., Streich, J.: Discretizing quantum field theories for quantum simulation (2020)
- [GMP+23] Grossman, P., Morrison, S., Penneys, D., Peters, E., Snyder, N.: The extended Haagerup fusion categories. *Ann. Sci. Éc. Norm. Supér. (4)* **56**(2), 589–664 (2023)
- [GNN09] Gelaki, S., Naidu, D., Nikshych, D.: Centers of graded fusion categories. *Algebra Number Theory* **3**(8), 959–990 (2009)
- [GNP21] Gong, Z., Nahum, A., Piroli, L.: Coarse-grained entanglement and operator growth in anomalous dynamics (2021)
- [GNVW09] Gross, D., Nesme, V., Vogts, H., Werner, R.: Index theory of one dimensional quantum walks and cellular automata. *Commun. Math. Phys.* **310**, 10 (2009)

- [GSSC20] Gong, Z., Sünderhauf, C., Schuch, N., Ignacio Cirac, J.: Classification of matrix-product unitaries with symmetries. *Phys. Rev. Lett.* **124**, 03 (2020)
- [Haa21] Haah, J.: Clifford quantum cellular automata: trivial group in 2D and Witt group in 3D. *J. Math. Phys.* **62**(9), 092202 (2021)
- [Haa22] Haah, J.: Topological phases of unitary dynamics: classification in Clifford category. [arXiv:2205.09141](https://arxiv.org/abs/2205.09141) (2022)
- [HJJY25] Hataishi, L., Jaklitsch, D., Jones, C., Yamashita, M.: On the structure of DHR bimodules of abstract spin chains (2025)
- [HK64] Haag, R., Kastler, D.: An algebraic approach to quantum field theory. *J. Math. Phys.* **5**, 848–861 (1964)
- [IW23] Inamura, K., Wen, X.: 2+1d symmetry-topological-order from local symmetric operators in 1+1d (2023)
- [JL24] Jones, C., Lim, J.: An index for quantum cellular automata on fusion spin chains. *Ann. Henri Poincaré* **25**(10), 4399–4422 (2024)
- [JM21] Jones, R.A., Metlitski, M.A.: One-dimensional lattice models for the boundary of two-dimensional majorana fermion symmetry-protected topological phases: Kramers-wannier duality as an exact Z_2 symmetry. *Phys. Rev. B* **104**, 245130 (2021)
- [JNP25] Jones, C., Naaijken, P., Penneys, D.: Holography for bulk-boundary local topological order (2025)
- [JNPW23] Jones, C., Naaijken, P., Penneys, D., Wallick, D.: Local topological order and boundary algebras. [arXiv: 2307.12552](https://arxiv.org/abs/2307.12552), (2023)
- [Jon83] Jones, V.: Index for subfactors. *Invent. Math.* **72**, 1–26 (1983)
- [Jon24] Jones, C.: DHR bimodules of quasi-local algebras and symmetric quantum cellular automata. *Quantum Topol.* (2024). [arXiv:2304.00068](https://arxiv.org/abs/2304.00068)
- [JP17] Jones, C., Penneys, D.: Operator algebras in rigid C^* -tensor categories. *Commun. Math. Phys.* **355**(3), 1121–1188 (2017)
- [JS97] Jones, V., Sunder, V.S.: Introduction to Subfactors. London Mathematical Society Lecture Note Series, Cambridge University Press (1997)
- [Kaw20] Kawahigashi, Y.: A remark on matrix product operator algebras, anyons and subfactors. *Lett. Math. Phys.* **110**(6), 1113–1122 (2020)
- [Kaw21] Kawahigashi, Y.: Two-dimensional topological order and operator algebras. *Int. J. Mod. Phys. B* **35**(8), 2130003–2616 (2021)
- [Kaw22] Kawahigashi, Y.: Projector matrix product operators, anyons and higher relative commutants of subfactors. *Math. Ann.* (2022)
- [KOZ22] Kaidi, J., Ohmori, K., Zheng, Y.: Symmetry TFTs for non-invertible defects. *Commun. Math. Phys.* **404**(2), 1021–1124 (2022)
- [KW41] Kramers, H.A., Wannier, G.H.: Statistics of the two-dimensional ferromagnet, Part I. *Phys. Rev.* **60**(3), 252–262 (1941)
- [LDOV23] Lootens, L., Delcamp, C., Ortiz, G., Verstraete, F.: Dualities in one-dimensional quantum lattice models: symmetric Hamiltonians and matrix product operator intertwiners. *PRX Quantum* **4**, 020357 (2023)
- [LE24] Long, D.M., Else, D.V.: Topological Phases of Many-Body Localized Systems: Beyond Eigenstate Order. [arXiv preprint](https://arxiv.org/abs/2401.12345) (2024)
- [LFH+20] Lootens, L., Fuchs, J., Haegeman, J., Schweigert, C., Verstraete, F.: Matrix product operator symmetries and intertwiners in string-nets with domain walls. *SciPost Phys.* (2020)
- [LR97] Longo, R., Roberts, J.: A theory of dimension. *K-Theory* **11**, 103–159 (1997)
- [LW05] Levin, M.A., Wen, X.G.: String-net condensation: A physical mechanism for topological phases. *Phys. Rev. B Condens. Matter Mater. Phys.* **71**, 045110 (2005). The OG
- [MLC24] Ma, R., Li, Y., Cheng, M.: Quantum cellular automata on symmetric subalgebras (2024)
- [MN18] Marshall, I., Nikshych, D.: On the Brauer-Picard groups of fusion categories. *Math. Z.* **288**(3–4), 689–712 (2018)
- [Naa17] Naaijken, P.: Quantum Spin Systems on Infinite Lattices, volume 933 of Lecture Notes in Physics. Springer, Cham (2017). A concise introduction
- [NS95] Nill, F., Szlachányi, K.: Quantum chains of Hopf algebras with quantum double cosymmetry. *Commun. Math. Phys.* **187**, 159–200 (1995)
- [NT13] Neshveyev, S., Tuset, L.: Compact quantum groups and their representation categories. *Cours Spécialisés [Specialized Courses]*, vol. 20. Société Mathématique de France, Paris (2013)
- [Ocn88] Ocneanu, A.: Quantized groups, string algebras and Galois theory for algebras. In: *Operator algebras and applications*, Vol. 2, volume 136 of London Math. Soc. Lecture Note Ser., pp. 119–172. Cambridge University Press, Cambridge (1988)
- [PC20] Piroli, L., Ignacio Cirac, J.: Quantum cellular automata, tensor networks, and area laws. *Phys. Rev. Lett.* **125**, 190402 (2020)

- [PFM+16] Po, H.C., Fidkowski, L., Morimoto, T., Potter, A.C., Vishwanath, A.: Chiral floquet phases of many-body localized bosons. *Phys. Rev. X* **6**, 041070 (2016)
- [PFVP17] Po, H.C., Fidkowski, L., Vishwanath, A., Potter, A.C.: Radical chiral floquet phases in a periodically driven Kitaev model and beyond. *Phys. Rev. B* **96**, 245116 (2017)
- [PM17] Potter, A.C., Morimoto, T.: Dynamically enriched topological orders in driven two-dimensional systems. *Phys. Rev. B* **95**, 155126 (2017)
- [PTSC21] Piroli, L., Turzillo, A., Shukla, S.K., Cirac, J.I.: Fermionic quantum cellular automata and generalized matrix-product unitaries. *J. Stat. Mech. Theory Exp.* **2021**(1), 013107 (2021)
- [PVF18] Potter, A.C., Vishwanath, A., Fidkowski, L.: Infinite family of three-dimensional floquet topological paramagnets. *Phys. Rev. B* **97**, 245106 (2018)
- [RWW20] Ranard, D., Walter, M., Witteveen, F.: A converse to Lieb-Robinson bounds in one dimension using index theory. *arXiv preprint [arXiv:2012.00741](https://arxiv.org/abs/2012.00741)* (2020)
- [Sch24] Schatz, K.: Boundary symmetries of (2+1)d topological orders. *arXiv preprint* (2024)
- [SSBC18] Sahinoglu, M.B., Shukla, S.K., Bi, F., Chen, X.: Matrix product representation of locality preserving unitaries. *Phys. Rev. B* **98**, 245122 (2018)
- [SW04] Schumacher, B., Werner, R.: Reversible quantum cellular automata. *arXiv Quantum Physics* (2004)
- [SWB+14] Sahinoglu, M.B., Williamson, D., Bultinck, N., Mariën, M., Haegeman, J., Schuch, N., Verstraete, F.: Characterizing topological order with matrix product operators. *Ann. Henri Poincaré* **22**(2), 563–592 (2014)
- [Tak79] Takesaki, M.: *Theory of operator algebras. I*. Springer, New York (1979)
- [TW19] Thorngren, R., Wang, Y.: *Fusion Category Symmetry I: Anomaly In-Flow and Gapped Phases*. *arXiv preprint* (2019)
- [TY98] Tambara, D., Yamagami, S.: Tensor categories with fusion rules of self-duality for finite abelian groups. *J. Algebra* **209**(2), 692–707 (1998)
- [VBW+18] Vanhove, R., Bal, M., Williamson, D.J., Bultinck, N., Haegeman, J., Verstraete, F.: Mapping topological to conformal field theories through strange correlators. *Phys. Rev. Lett.* **121**, 177203, 10 (2018)
- [Wat95] Watrous, J.: On one-dimensional quantum cellular automata. In: *Proceedings of IEEE 36th Annual Foundations of Computer Science*, pp. 528–537 (1995)
- [WBV17] Williamson, D.J., Bultinck, N., Verstraete, F.: Symmetry-enriched topological order in tensor networks: defects, gauging and anyon condensation. *arXiv preprint* (2017)
- [ZCZW19] Zeng, B., Chen, X., Zhou, D., Wen, X.: *Quantum information meets quantum matter*. Quantum Science and Technology. Springer, New York (2019). From quantum entanglement to topological phases of many-body systems, With a foreword by John Preskill
- [ZL21] Zhang, C., Levin, M.: Classification of interacting floquet phases with $u(1)$ symmetry in two dimensions. *Phys. Rev. B* **103**, 064302 (2021)

Communicated by Y. Kawahigashi